Biochemistry WQE

Detailed Study Guide • BMED 795 • 2026 outline

Prepared from the supplied exam outline and focused notes

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Contents

Scope, sources, and how to study 2

1. Chemical foundations: water, pH, buffers, and energy 4

2. Amino acids and physical properties of proteins 5

3. Protein secondary, tertiary, and quaternary structure 7

4. Myoglobin and hemoglobin 8

5. Antibodies and antigens 9

6. Carbohydrate structure and chemistry 10

7. Membrane organization 12

8. Membrane fluidity and transport 12

9. Glycolysis, fermentation, pyruvate oxidation, and the TCA cycle 14

10. Gluconeogenesis and reciprocal control 16

11. Oxidative phosphorylation 17

12. Fatty acid oxidation 19

13. Enzyme catalysis, kinetics, and inhibition 20

14. Hormone signaling and metabolic integration 21

15. Caloric restriction and aging 23

16. G-protein signaling 24

17. Membrane receptor signaling and pathway comparison 25

18. Integration tools and supplementary concepts 28

19. Original practice paper 30

20. Worked answers and marking guide 32

21. Revision schedule and final readiness check 34

Source register 35

Scope, sources, and how to study

This guide follows **all 17 topics** in the supplied *05_Biochemistry-BMEB_2026.docx*. That outline describes the BMED 795 PhD Written Qualifying Examination as **closed book and closed notes**, with a **basic scientific calculator** and **one A4 sheet, front and back, containing formulas only**. Treat this guide as preparation material, not material permitted in the examination room. The separate formula-sheet PDF contains equations only; confirm any updated exam rules with the department.

Source key. [O] means the supplied exam-outline DOCX, especially its topic/reading table. **[F, pp. x–y]** means pages of the supplied 58-page *Biochemistry_WQE_Focused_Study_Guide.pdf*; its printed page numbers match PDF pages. Section source notes identify the supplied coverage. Detailed explanations, additional examples, the practice exam, and its answers are newly written educational material.

Textbook limitation. The uploaded *Principles of Biochemistry*, fifth edition, could not be opened because its file exceeds the available 32 MiB transfer limit. Chapter assignments below are reproduced from [O], not independently verified against that PDF. Sections absent or brief in [F] are expanded using standard biochemistry knowledge and are identified as such. No textbook page numbers, quotations, or claims of textbook verification are invented. The syllabus names Moran, Horton, Scrimgeour, and Perry as the primary authors.

Depth and priorities. All listed topics are required scope; no official marks distribution or past paper was supplied. Emphasis here is a study recommendation, not an exam prediction. First master acid–base calculations, protein structure/function, enzyme kinetics, pathway accounting, oxidative phosphorylation, and signal-to-metabolic-response reasoning. Then use comparisons and drawing practice to strengthen the remaining topics. Nitrogen metabolism and some laboratory methods appear in [F] but are not separate topics in [O]; they are supplementary material.

Exam-alignment map

Official topic	Guide section	Chapters assigned in [O]	Supplied focused-guide coverage
01. Introduction; water, buffers, pH	1	1–2	pp. 3–9, 25–26
02. Amino acids; physical properties of proteins	2	3–4	pp. 10–15, 42–43
03. Secondary, tertiary, quaternary structure	3	4	pp. 14–15
04. Hemoglobin and myoglobin	4	4	pp. 16–17
05. Antibodies and antigens	5	4	ELISA only, p. 42; expanded here
06. Carbohydrates	6	7	pp. 27–34
07. Membranes	7	9	pp. 35–38
08. Fluidity and transport	8	9	pp. 35–41, 44
09. Glycolysis, fermentation, citric acid cycle	9	11, 13	pp. 45–50, 57
10. Gluconeogenesis	10	12	Brief pyruvate-carboxylation mention, p. 47; expanded here
11. Oxidative phosphorylation	11	14	pp. 51–53
12. Fatty acid oxidation	12	16	pp. 54–55
13. Enzymes: kinetics and inhibitors	13	5	pp. 18–26
14. Hormone signaling	14	10, 12, 16	General phosphorylation, p. 24; expanded here

Official topic	Guide section	Chapters assigned in [O]	Supplied focused-guide coverage
15. Caloric restriction and aging	15	16	No dedicated coverage; expanded here
16. G-protein signaling	16	9	No dedicated coverage; expanded here
17. Membrane receptor signaling	17	9	No dedicated coverage; expanded here

A productive study cycle

1. **Retrieve before reading.** Write the pathway, comparison, or mechanism from memory, then correct it.
2. **Explain causally.** For each perturbation write: initial change → molecular mechanism → pathway effect → physiological consequence.
3. **Calculate with units.** State the reaction basis, compartments, ATP-yield convention, and assumptions before substituting values.
4. **Draw.** Practice amino acid classes, a peptide, an α-helix/β-sheet sketch, glucose anomers, an antibody, a bilayer, respiratory electron flow, and the main signaling cascades.
5. **Transfer.** Ask what changes if oxygen disappears, a transporter is inhibited, an enzyme is phosphorylated, or a receptor cannot turn off.

For written answers, start with the direct answer, explain the mechanism, give the expected observation, and state a relevant limitation. A clear chain of reasoning is stronger than a list of loosely connected terms.

1. Chemical foundations: water, pH, buffers, and energy

You should be able to: calculate concentrations and buffer changes; explain weak interactions; distinguish reaction favorability from rate. **Sources:** [O], topic 01; [F, pp. 3–9, 25–26].

Water and noncovalent interactions

Water is polar and forms an extensive hydrogen-bond network. Its high dielectric constant reduces electrostatic attraction between dissolved charges relative to a nonpolar environment. Hydration shells orient oxygen toward cations and hydrogen toward anions. This supports dissolution of many ions and polar molecules.

Interaction	Molecular basis	Biological importance and qualification
Hydrogen bond	Donor H interacts with an acceptor electronegative atom	Directional; backbone structure, ligand recognition, water networks
Electrostatic interaction	Attraction or repulsion between charges	Sensitive to pH, salt, dielectric environment, and distance
van der Waals interaction	Short-range attractions from fluctuating dipoles	Many contacts stabilize closely packed protein interiors; overlap causes repulsion
Hydrophobic effect	Nonpolar groups cluster in water, changing solvent organization	Major driver of folding and bilayer assembly; not a new covalent bond
Disulfide bond	Oxidation of two cysteine thiols	Covalent stabilization, especially in many secreted proteins

Hydrophobic burial usually stabilizes a soluble protein, but a buried polar group also needs favorable interactions. The hydrophobic effect involves solvent entropy and enthalpy; do not reduce every folding event to an entropy increase of the protein itself. Folding decreases the chain's conformational entropy.

Concentration and acid–base equations

$$n = \frac{m}{M}, \qquad C = \frac{n}{V}, \qquad C_1V_1 = C_2V_2$$

Here M is molar mass, in g mol^{-1}; relative molecular mass M_r is formally dimensionless. Volume must match the concentration units.

$$K_a = \frac{[H^+][A^-]}{[HA]}, \quad pK_a = -\log_{10} K_a, \quad pH = -\log_{10} a_{H^+}$$

Concentrations approximate activities in typical dilute teaching problems. At 25 °C, $K_w \approx 10^{-14}$ and $pH + pOH \approx 14$; those numbers depend on temperature.

$$pH = pK_a + \log_{10} \frac{[A^-]}{[HA]}$$

At $pH = pK_a$, acid and conjugate base are equal. One unit above pK_a gives base:acid = 10:1; one unit below gives 1:10. A practical buffering range is approximately $pK_a \pm 1$. Buffer capacity increases with total buffer concentration and is greatest near pK_a. Dilution can preserve the ratio and approximately preserve pH while reducing capacity.

When adding strong acid or base: perform neutralization in moles first, then apply Henderson–Hasselbalch to the remaining conjugate pair. Do not put the original concentrations into the equation. If one component is exhausted, calculate excess strong acid/base or the relevant weak-acid equilibrium instead.

Worked example 1: adding acid to a buffer

A 100 mL buffer contains 10.0 mmol HA and 20.0 mmol A^-, with $pK_a = 6.00$. Initial pH = $6.00 + \log_{10} 2 = 6.30$. Add 5.00 mmol HCl:

$$A^- + H^+ \rightarrow HA$$

New amounts: A^- = 15.0 mmol; HA = 15.0 mmol. Therefore pH = 6.00. The common final volume cancels in the ratio. This treatment assumes the added acid dissociates fully and the conjugate pair remains a suitable buffer.

Biological buffers

- **Bicarbonate/CO_2:** important extracellularly; lungs regulate CO_2 and kidneys regulate acid/base handling. At approximately 37 °C, blood problems often use $pH = 6.1 + \log_{10}([HCO_3^-]/(0.03P_{CO_2}))$, with bicarbonate in mmol/L and pressure in mmHg. Use the constants supplied in a question.
- **Phosphate:** relevant inside cells and in renal tubular fluid.
- **Proteins:** terminal groups and ionizable side chains buffer; histidine is particularly useful because its protonation can change near physiological conditions. Its effective pK_a depends on its environment.

Reaction energy and coupling

$$\Delta G = \Delta G^{\circ\prime} + RT \ln Q, \qquad \Delta G^{\circ\prime} = -RT \ln K'_{eq}$$

The prime indicates a biochemical transformed standard state, conventionally fixing pH. Use natural logarithms here; Henderson–Hasselbalch uses base 10. At equilibrium, $\Delta G = 0$, not necessarily $\Delta G^{\circ\prime} = 0$. A favorable reaction may be slow if its activation barrier is high. Enzymes lower that barrier, not the net reaction free energy or equilibrium constant.

For linked reactions, free energies add: $\Delta G_{total} = \sum_i \Delta G_i$. ATP hydrolysis can drive an unfavorable step only when the chemistry actually couples them. In a redox reaction, $\Delta G = -nF\Delta E$, where n is the number of electrons transferred and $\Delta E = E_{acceptor} - E_{donor}$ for consistently defined reduction potentials. A positive potential difference gives a favorable negative free-energy change.

Worked example 2: at 298 K, let $\Delta G^{\circ\prime} = +5.0$ kJ/mol and $Q = 0.010$. With $R = 0.008314$ kJ mol^{-1} K^{-1}, $\Delta G = 5.0 + (0.008314)(298) \ln 0.010 = -6.41$ kJ/mol. Cellular reactant/product ratios can make a standard-state unfavorable reaction favorable.

Check yourself: Why can a concentrated and dilute buffer have the same pH? Why does catalysis not change equilibrium? Can a reaction with positive standard free energy proceed forward? Answer each by referring to the governing variable, not just a memorized slogan.

2. Amino acids and physical properties of proteins

You should be able to: classify all 20 side chains, estimate charge and pI, explain peptide chemistry, and choose a separation method. **Sources:** [O], topic 02; [F, pp. 10–15, 42–43]. Classification and quantitative examples are expanded here.

Side-chain reference

Class	Amino acids; three-letter / one-letter codes	Essential structural point
Mainly nonpolar aliphatic	Gly G, Ala A, Val V, Leu L, Ile I, Met M, Pro P	Gly is achiral and flexible; Pro restricts backbone geometry; Met has a thioether

Class	Amino acids; three-letter / one-letter codes	Essential structural point
Aromatic	Phe F, Tyr Y, Trp W	Tyr also has a polar phenol; Trp and Tyr dominate protein absorbance near 280 nm
Polar, usually uncharged	Ser S, Thr T, Asn N, Gln Q, Cys C	Hydroxyls, amides, or thiol; Cys can form disulfides
Acidic	Asp D, Glu E	Usually negatively charged near pH 7
Basic	Lys K, Arg R, His H	Lys/Arg usually positive; His often mainly neutral but readily exchanges protons

Learn the actual structures: Asn/Gln are side-chain amides, not negatively charged versions of Asp/Glu. Ser, Thr, and Tyr are common phosphorylation sites. Cys can be a catalytic nucleophile. Proline's backbone nitrogen usually lacks an N–H donor in a peptide.

Approximate free-residue ionization values: α-carboxyl 2; Asp 3.9; Glu 4.2; His 6.0; Cys 8.3; α-amino 9–10; Tyr 10.1; Lys 10.5; Arg 12.5. **These are study approximations, not universal constants.** Protein microenvironments shift values; use supplied data in calculations.

Net charge and isoelectric point

An acidic group is neutral when protonated and negative when deprotonated. A basic group is positive when protonated and neutral when deprotonated. For a peptide count **one free N-terminus, one free C-terminus, and all ionizable side chains**. Internal peptide bonds do not contribute ordinary free α-amino/α-carboxyl titrations.

$$f_{A^-} = \frac{1}{1 + 10^{pK_a - pH}}, \qquad f_{BH^+} = \frac{1}{1 + 10^{pH - pK_a}}$$

Multiply each fraction by its group's charge and sum. For rough estimates far from a pK_a, assign integer charges. At a group's pK_a, it is half protonated.

The **pI** is the pH at which average net charge is zero. For a simple amino acid with a well-defined neutral form between two titrations, average the **two pK_a values that flank that form**. Do not blindly average the smallest and largest values or always use the terminal groups. For complex peptides, solve the charge balance or identify the appropriate titration region carefully.

Worked example 3: glutamate. Given pK_a values 2.2, 4.3, and 9.7, net charges progress $+1 \to 0 \to -1 \to -2$ as pH rises. The neutral form lies between 2.2 and 4.3, so $pI = (2.2 + 4.3)/2 = 3.25$. At pH 7 it is mainly -1.

Worked example 4: peptide Lys-Gly-Asp at pH 7. Approximate contributions: N-terminus +1; Lys side chain +1; Asp side chain −1; C-terminus −1. Total ≈ 0. This does not establish that the exact pI is 7; fractional charges and actual pK_a values matter.

Peptide bond and protein behavior

Peptide sequences run N → C. Resonance gives the peptide bond partial double-bond character, so its atoms are approximately planar and rotation is restricted. Most peptide bonds are trans; bonds before proline have a greater cis fraction than most others. Backbone φ and ψ torsion angles provide much of the conformational freedom. Steric constraints underlie the Ramachandran plot.

At pH below pI a protein tends to be net positive and migrates toward the **cathode**; above pI it tends to be negative and moves toward the **anode**. Solubility is often lower near pI because net electrostatic repulsion is reduced, although protein-specific behavior matters.

Method	Separation/detection basis	Typical exam trap
Size-exclusion chromatography	Hydrodynamic size	Larger species generally elute first; shape and oligomerization affect apparent size
Ion-exchange chromatography	Interaction with charged resin	Cation exchangers bind positively charged proteins; anion exchangers bind negative ones
SDS-PAGE	Approximate polypeptide mass after denaturation and SDS binding	Reducing agents additionally break disulfides; native oligomer size is not directly measured
Isoelectric focusing	pI in a pH gradient	Proteins focus where net charge is zero
Affinity chromatography	Specific reversible binding	Binding partner must be appropriate to the target

Check yourself: Why does a reducing SDS gel separate some antibody chains that a nonreducing gel retains together? Why is charge-based chromatography sensitive to buffer pH?

3. Protein secondary, tertiary, and quaternary structure

You should be able to: distinguish structural levels, explain stabilizing interactions, and predict consequences of sequence changes. **Sources:** [O], topic 03; [F, pp. 14–15]. Structural detail and applications are expanded here.

Level	Definition	What to draw or explain
Primary	Amino acid sequence and covalent connectivity	N → C chain; disulfide connectivity when specified
Secondary	Local recurring backbone arrangements	α-helix, β-sheet, turns; backbone hydrogen bonds
Tertiary	Complete fold of a single polypeptide	Core, surface, domains, active/binding site
Quaternary	Arrangement of multiple polypeptides	Subunits, interfaces, symmetry, cooperative changes

An **α-helix** is usually right-handed for L-amino-acid proteins, has about 3.6 residues per turn, and uses backbone C=O(i)···H–N($i+4$) hydrogen bonds. Side chains project outward. Proline often disrupts a regular helix; glycine's flexibility can also destabilize it, but neither rule is absolute.

β-sheets contain extended strands with backbone hydrogen bonds between strands. Strands can be parallel or antiparallel; side chains alternate above and below the sheet. Do not draw all side chains on the same face. Turns connect secondary elements, and many functional regions are flexible rather than regularly helical or sheet-like.

In a soluble globular protein, nonpolar side chains are usually buried. In a membrane-spanning segment, nonpolar side chains often face the lipid. Secondary-structure hydrogen bonds help satisfy polar backbone groups within that hydrophobic environment. Disulfides add stability to many extracellular proteins; the reducing cytosol generally disfavors stable disulfide formation.

Folding, denaturation, and mutations

Folding balances chain entropy, solvent effects, packing, hydrogen bonds, and electrostatics. Chaperones assist folding and prevent aggregation; they do not supply a new amino acid sequence. Denaturation commonly disrupts secondary, tertiary, and quaternary structure while leaving peptide bonds intact. Refolding may be possible after removing the denaturant, but aggregation or covalent damage can prevent recovery.

Prediction template: locate the residue → identify its original interaction → determine what the replacement changes → predict effects on stability or function → acknowledge what cannot be concluded without data.

Example: replacing a buried leucine with aspartate can introduce a poorly solvated charge and destabilize the fold. Replacing a surface leucine may have a smaller structural effect. A conservative substitution at an active site can nevertheless have a large functional effect. Location matters as much as residue class.

Example: a tetramer held only by noncovalent interactions can yield one subunit band on denaturing SDS-PAGE. If interchain disulfides connect subunits, reducing conditions may be needed for full chain separation. A single band does not prove that the native protein is monomeric.

Check yourself: Distinguish a domain from a subunit. A domain is a folding/functional unit within a polypeptide; a subunit is a separate polypeptide in a complex. Explain why unfolding can abolish activity without breaking the primary sequence.

4. Myoglobin and hemoglobin

You should be able to: interpret oxygen-binding curves, explain cooperativity and allostery, and predict changes in tissue delivery. **Sources:** [O], topic 04; [F, pp. 16–17]. Quantitative binding and additional examples are expanded here.

Both proteins bind oxygen through heme containing **ferrous iron, Fe^{2+}**. Oxidation to Fe^{3+} produces methemoglobin, which does not bind oxygen normally at that site. Oxygen binding is not equivalent to permanently oxidizing the heme iron to the ferric state.

Feature	Myoglobin	Adult hemoglobin A
Main role	Muscle oxygen storage and facilitated diffusion	Blood oxygen transport
Structure	One chain, one heme	$\alpha_2\beta_2$ tetramer, four hemes
Binding	Hyperbolic, noncooperative	Sigmoidal, cooperative
Typical affinity	Relatively high	Changes with allosteric state and conditions
Major regulation	No Hb-like quaternary cooperativity	H^+, CO_2, 2,3-BPG, temperature

Oxygen binding favors Hb's higher-affinity **R state** over the lower-affinity **T state**. T/R terminology describes conformational ensembles; it does not require every subunit in an individual molecule always to have identical ligand occupancy. Cooperativity means a ligand-binding event changes the affinity of other sites.

$$Y_{Mb} = \frac{pO_2}{P_{50} + pO_2}, \qquad Y_{Hill} = \frac{(pO_2)^{n_H}}{P_{50}^{n_H} + (pO_2)^{n_H}}$$

Y is fractional saturation and P_{50} is the oxygen pressure giving 50% saturation. **Lower P_{50} means higher affinity.** The Hill equation is a useful empirical approximation. $n_H > 1$ suggests positive cooperativity; its fitted value is not simply the number of subunits.

Physiological shifts

- **Right shift:** increased H^+, CO_2, temperature, or 2,3-BPG generally favors oxygen unloading; P_{50} rises.
- **Bohr effect:** H^+ and CO_2 reduce Hb oxygen affinity through linked protonation/carbamate chemistry and stabilization of lower-affinity conformations. Active tissues thereby favor unloading.
- **2,3-BPG:** binds preferentially in the central cavity of deoxygenated adult Hb and stabilizes the T state.
- **Fetal Hb:** has γ rather than β chains and binds 2,3-BPG less strongly; it has higher oxygen affinity than adult Hb under comparable conditions, supporting placental transfer.
- **Carbon monoxide:** occupies oxygen-binding sites and increases affinity at remaining sites, impairing unloading as well as reducing capacity. A left shift does not mean improved delivery.

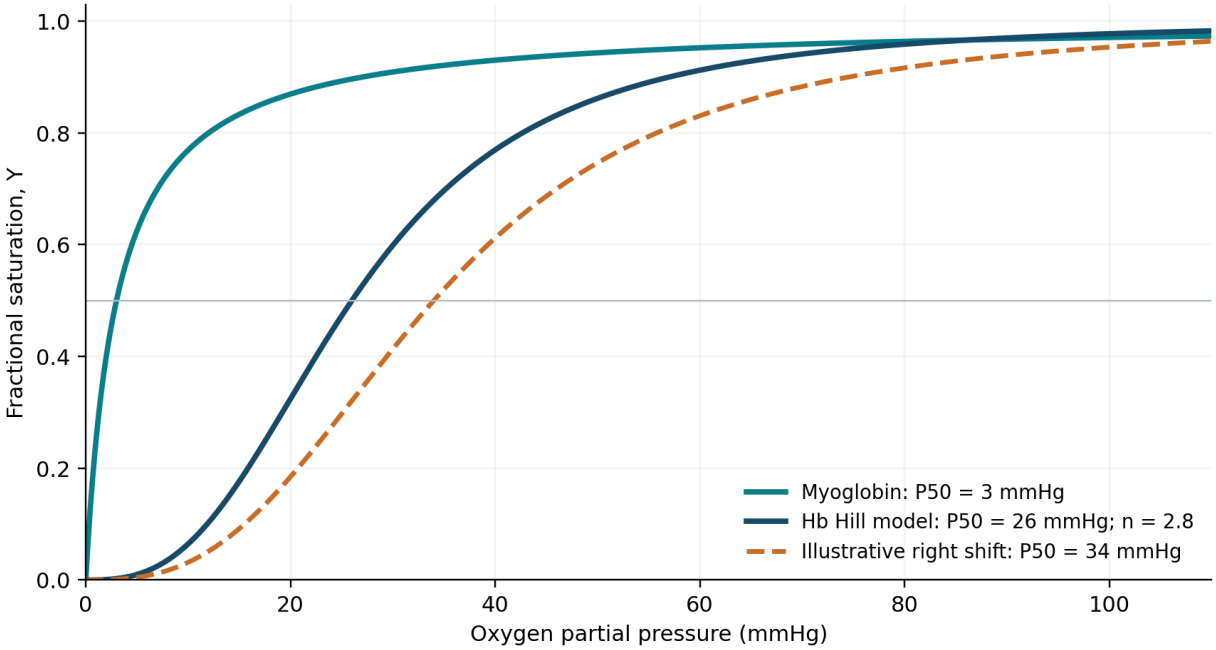


Figure 1: Illustrative oxygen-binding models: the right-shifted Hb curve has lower affinity at a given oxygen pressure. Curves are teaching models, not clinical reference data.

Worked example 5: for a hypothetical Hb model with $P_{50} = 26$ mmHg and $n_H = 2.8$, Y at 100 mmHg ≈ 0.978 and at 40 mmHg ≈ 0.770. The modeled saturation difference is ≈ 0.208, or 20.8 percentage points. For a hypothetical Mb with $P_{50} = 3$ mmHg, Y at 40 mmHg is $40/43 = 0.930$. These are illustrative model values, not patient predictions.

Check yourself: Why does a sigmoidal curve support both loading and unloading? Why is oxygen affinity different from total oxygen-carrying capacity? Explain why severe anemia can impair delivery despite high saturation.

5. Antibodies and antigens

You should be able to: label an antibody, distinguish affinity/avidity and antigen/epitope, and interpret a simple immunoassay. **Sources:** [O], topic 05; [F, p. 42] for ELISA only. Most of this section is standard-knowledge expansion, not textbook-verified coverage.

An **antigen** is a molecule or structure recognized by an immune receptor; an **epitope** is the specific part recognized. An **immunogen** elicits an immune response. Recognition and ability to elicit a response are related but not identical properties. A hapten may bind an antibody yet require a carrier to elicit a strong response.

Antibody architecture

A conventional IgG consists of **two identical heavy chains and two identical light chains**, joined by disulfides and noncovalent interactions. Each chain has variable and constant domains built on the immunoglobulin β-sandwich fold. Variable regions of one heavy and one light chain combine to create each antigen-binding site; complementarity-determining regions contribute prominently to specificity.

- **Fab regions:** two antigen-binding arms in a conventional IgG.
- **Fc region:** heavy-chain constant region involved in effector interactions, including Fc receptors and, for appropriate classes/subclasses, complement.
- **Hinge:** permits flexibility; antigen geometry affects whether both arms bind effectively.
- **Linear epitope:** a contiguous sequence; it may remain detectable after denaturation.
- **Conformational epitope:** assembled by folding; denaturation can destroy recognition.

Binding involves complementary shape and chemistry: hydrogen bonds, electrostatics, van der

Waals contacts, and hydrophobic interactions. Antigen binding is normally noncovalent and reversible; antibody disulfides stabilize the antibody rather than forming a covalent antibody–antigen bond.

$$K_d = \frac{[Ab][Ag]}{[Ab{:}Ag]}, \qquad \theta = \frac{[Ag]}{K_d + [Ag]}$$

Lower K_d means higher **intrinsic affinity** for a single interaction. The occupancy equation assumes one class of independent binding sites and uses free antigen concentration. **Avidity** is overall multivalent binding strength, influenced by affinity, valency, and geometry. IgM can have high avidity even when individual-site affinity is modest.

Isotype	Recognition-level facts
IgG	Major serum antibody; many effector functions; placental transfer
IgM	Usually pentameric when secreted; early response; high multivalent avidity
IgA	Important at mucosal surfaces; secretory form usually dimeric
IgE	Allergy and antiparasite responses through high-affinity receptors on effector cells
IgD	Mainly a B-cell surface receptor role

Monoclonal antibodies recognize a particular epitope through one clone's binding specificity; **polyclonal** preparations contain multiple specificities against an antigen. Do not infer perfect absence of cross-reactivity from "monoclonal."

Assay reasoning

A sandwich ELISA commonly uses an immobilized capture antibody, sample antigen, and a detection antibody recognizing an accessible second epitope. The enzyme label may be on the detection antibody or on an additional reagent. Washing reduces unbound material; substrate conversion generates a measured signal. Within the validated calibration range, greater target often gives greater signal. **Competitive** assays can show the opposite relationship.

Use a blank, positive and negative controls, and a standard curve. At very high antigen levels some sandwich assays can show a high-dose hook effect; dilution checks can help. Cross-reactivity, poor washing, damaged epitopes, and matrix effects can distort results.

Worked example 6: with $K_d = 10$ nM and free antigen = 30 nM, occupancy is $30/(10+30) = 0.75$. A second antibody with $K_d = 1$ nM has greater intrinsic affinity, but that alone does not determine every multivalent assay outcome.

Check yourself: Why can an antibody work in ELISA but fail on a denaturing Western blot? Which parts bind antigen and which mediate effector functions? Why must the capture and detection antibodies in a sandwich assay avoid mutually exclusive binding?

6. Carbohydrate structure and chemistry

You should be able to: classify sugars, draw and distinguish stereoisomers, identify reducing ends, and connect linkage to function. **Sources:** [O], topic 06; [F, pp. 27–34].

Classification and stereochemistry

An **aldose** has an aldehyde in its open-chain form; a **ketose** has a ketone. Glucose is an aldohexose, fructose a ketohexose, and ribose an aldopentose. D/L configuration is determined by the highest-numbered stereocenter in the Fischer projection relative to glyceraldehyde, **not by the sign of optical rotation**.

Relationship	Definition	Example
Enantiomers	Nonsuperimposable mirror images	D-glucose and L-glucose
Epimers	Differ at one stereocenter, within the same connectivity	D-glucose/D-mannose at C2; D-glucose/D-galactose at C4
Anomers	Differ at the anomeric center formed on cyclization	α-D-glucopyranose and β-D-glucopyranose
Constitutional isomers	Different connectivity	Open-chain glucose and fructose

Cyclization of an aldose forms an intramolecular **hemiacetal**; a ketose forms a **hemiketal**. The original carbonyl carbon becomes the anomeric carbon: C1 in glucose, C2 in fructose. A pyranose ring has six ring atoms; a furanose ring has five. Ring size is not the total number of carbons in the molecule.

In standard Haworth drawings of D-glucopyranose with the reference CH_2OH above the ring, α has the anomeric OH below and β above. More generally, α/β describes the relationship to the reference substituent, so do not apply "α always down" to every sugar/drawing. **Mutarotation** is interconversion through ring opening and reclosing, accompanied by a change in optical rotation.

Drawing drill: glucose

Draw D-glucose in Fischer form with C2–C5 OH positions **right, left, right, right**. For the conventional D-glucopyranose Haworth orientation: groups on the Fischer right become down and those on the left become up; CH_2OH is up. Place the C1 OH down for α and up for β. This shortcut depends on maintaining that conventional orientation. In the common $^4C_1$ chair of β-D-glucopyranose the substituents are favorably arranged largely equatorially.

Reducing behavior and linkages

A free anomeric hemiacetal/hemiketal can equilibrate with an open-chain carbonyl form and permits reducing behavior under appropriate test conditions. Ketoses such as fructose can also give positive common alkaline reducing-sugar tests through tautomerization. Do not identify all ketoses as nonreducing.

Carbohydrate	Linkage/structure	Reducing behavior or function
Maltose	Glc-α(1→4)-Glc	One free anomeric center; reducing
Lactose	Gal-β(1→4)-Glc	One free anomeric center; reducing
Sucrose	Glc-α(1↔2)-β-Fru	Both anomeric centers linked; nonreducing
Amylose	Glucose α(1→4)	Mainly unbranched plant storage component
Amylopectin	α(1→4), α(1→6) branches	Branched plant storage component
Glycogen	α(1→4), frequent α(1→6) branches	Animal storage; many nonreducing ends support rapid mobilization
Cellulose	Glucose β(1→4)	Structural fibers; humans lack cellulase

An ordinary glycogen particle has many nonreducing ends and one core-associated reducing end, usually attached to glycogenin. Branching increases the number of sites for synthesis and breakdown; it does not create many independent reducing ends.

Chemical reactions emphasized in the supplied notes

- **Mild C1 oxidation of glucose:** gluconic acid, an aldonic acid. Oxidation at terminal C6 instead yields a uronic acid; oxidation at both ends yields an aldaric acid under appropriate conditions.

- **Reduction of the carbonyl:** produces a sugar alcohol; glucose gives sorbitol.
- **Glycoside formation:** acid-catalyzed reaction of an anomeric OH with an alcohol makes an acetal/ketal glycoside and fixes that center until hydrolysis. A methyl glycoside no longer mutarotates at that locked center.
- **Kiliani–Fischer extension:** lengthens an aldose by one carbon and yields two products epimeric at the new C2. This reaction is in [F]; its exam weighting is unknown.

Check yourself: Draw both glucose anomers and identify C1. Explain why lactose reduces a test reagent but sucrose does not. Why can humans digest starch but not cellulose despite both being glucose polymers?

7. Membrane organization

You should be able to: explain bilayer assembly, lipid asymmetry, membrane protein placement, and lipid-class functions. **Sources:** [O], topic 07; [F, pp. 35–38].

Amphipathic lipids have polar head groups and nonpolar tails. Bilayer assembly reduces exposure of hydrophobic tails to water. Bilayers are fluid, asymmetric, selectively permeable structures rather than rigid walls. Small defects tend to reseal because exposed hydrophobic edges are energetically unfavorable.

Lipid class	Core structure	Main connection
Triacylglycerol	Glycerol esterified to three fatty acids	Neutral storage lipid, not a principal bilayer-forming lipid
Glycerophospholipid	Glycerol, two acyl chains, phosphate-containing head	Major bilayer component; head-group chemistry matters
Sphingomyelin	Sphingosine-based backbone, amide-linked fatty acid, phosphocholine	Membrane lipid prominent in many animal membranes
Glycolipid	Lipid carrying carbohydrate	Recognition and surface organization
Cholesterol	Four-ring sterol with small polar OH	Modulates packing/permeability; precursor of steroids, bile acids, vitamin D

Lateral diffusion is generally much faster than spontaneous transbilayer flip-flop of phospholipids because a polar head crossing the hydrophobic core is unfavorable. Flippases/floppases and scramblases regulate distribution; their mechanisms and energy dependence differ. Phosphatidylserine is usually enriched in the cytosolic leaflet of a healthy plasma membrane, and its external exposure can signal apoptosis.

Integral proteins enter the hydrophobic core, often through transmembrane α-helices; some membranes contain β-barrel proteins. **Peripheral proteins** associate with surfaces or other proteins. Lipid anchors provide another attachment route. Glycans on plasma membrane glycoproteins and glycolipids face the extracellular side; the corresponding orientation in the secretory pathway is luminal.

Membrane permeability depends on solute size, polarity, charge, and lipid solubility. O_2 and CO_2 cross readily; ions usually need proteins. Water crosses to some extent through lipid but aquaporins greatly increase water permeability. A membrane protein's hydrophobic span and orientation must match the bilayer environment.

Check yourself: Why do triacylglycerols form storage droplets rather than ordinary phospholipid bilayers? Why does transbilayer lipid motion require more assistance than lateral diffusion? Why is a protein's cytosolic glycosylation pattern different from its extracellular glycan display?

8. Membrane fluidity and transport

You should be able to: predict fluidity changes, classify transport, calculate an electrochemical driving force, and reason about osmosis. **Sources:** [O], topic 08; [F, pp. 35–41, 44]. Quantitative transport is expanded here.

Fluidity

At a fixed temperature, **shorter chains** and **more cis double bonds** usually reduce packing and increase fluidity. Longer saturated chains generally raise melting temperature. Trans unsaturated chains pack more like saturated chains than comparable cis chains. Cholesterol restrains acyl-chain motion at high temperature and disrupts tight packing at low temperature; it buffers fluidity and often reduces permeability. Its effect depends on composition and temperature.

Transport mechanisms

Mechanism	Driving force	Key distinction	Example
Simple diffusion	Down chemical potential	No transporter; rate often scales with gradient	O_2 across membrane
Channel-mediated diffusion	Down electrochemical gradient	High-throughput pore; often gated	K^+ channel, aquaporin
Carrier-mediated facilitated diffusion	Down electrochemical gradient	Alternating access; binding and saturation	GLUT glucose transporter
Primary active transport	Direct energy coupling	Can move substrate uphill	Na^+/K^+-ATPase
Secondary active transport	Downhill movement of another solute	Coupled gradients, not direct ATP hydrolysis by the cotransporter	Na^+/glucose symporter
Vesicular transport	Energy-dependent membrane trafficking	Bulk cargo and receptor-dependent routes	Endocytosis/exocytosis

"Uses a protein" does not mean active transport. **Symport** means coupled solutes move in the same direction; **antiport** means opposite directions. Those words alone do not specify the energy source. Carriers display saturability; channels are not simply unsaturable carriers and have their own gating/conductance properties.

Na^+/K^+-ATPase typically exports **3 Na^+** and imports **2 K^+ per ATP**, producing a net outward positive charge and maintaining gradients. A Na^+/glucose cotransporter uses the Na^+ gradient; inhibiting the pump can eventually impair glucose uptake as that gradient dissipates. The effect need not be instantaneous.

Electrochemical calculations

For transfer from side 1 to side 2:

$$\Delta G_{1\to 2} = RT \ln \frac{C_2}{C_1} + zF(\psi_2 - \psi_1)$$

Here z is ion charge and $F = 96485\ \text{C mol}^{-1}$. Use volts, kelvin, and consistent joule/kilojoule units. Negative ΔG favors that direction. Neutral solutes have $z = 0$. At electrochemical equilibrium:

$$\psi_2 - \psi_1 = \frac{RT}{zF} \ln \frac{C_1}{C_2}$$

Worked example 7: Na^+ entry. At 310 K, outside Na^+ = 140 mM, inside = 14 mM, and inside potential relative to outside = −0.060 V. For outside → inside:

$$\Delta G = (8.314)(310)\ln(14/140) + (1)(96485)(-0.060)$$

The chemical term is −5.93 kJ/mol, electrical term −5.79 kJ/mol, total **−11.72 kJ/mol**. Both gradients favor entry. That free energy can help drive another solute uphill when total coupled ΔG remains negative.

Osmosis and vesicular routes

Water moves toward lower water chemical potential, often toward the side with greater effective nonpenetrating solute concentration. **Osmolarity** counts dissolved particles; **tonicity** predicts sustained cell-volume effects and depends on membrane permeability. A permeating solute can contribute to osmolarity without producing the same lasting tonicity as an impermeant one. For dilute solutions, $\pi \approx iCRT$.

Endocytosis internalizes membrane and cargo. Phagocytosis engulfs large particles; receptor-mediated endocytosis concentrates specific ligands, such as LDL binding its receptor. Exocytosis releases cargo and adds vesicle membrane to the cell surface.

Check yourself: Can a cation enter against its concentration gradient without ATP hydrolysis? Yes, if the electrical term makes total ΔG negative. Explain why blocking a primary pump can inhibit a secondary transporter.

9. Glycolysis, fermentation, pyruvate oxidation, and the TCA cycle

You should be able to: reconstruct the pathways, identify carbon/energy changes, distinguish irreversible control steps, and connect oxygen to NAD^+ regeneration. **Sources:** [O], topic 09; [F, pp. 45–50, 57]. Reaction-by-reaction detail is expanded here.

Glycolysis: cytosol, one glucose to two pyruvate

Step	Enzyme and conversion	Accounting or control
1	Hexokinase/glucokinase: glucose → glucose-6-phosphate	Uses ATP; effectively irreversible
2	Phosphoglucose isomerase: G6P → fructose-6-phosphate	Aldose/ketose rearrangement
3	PFK-1: F6P → fructose-1,6-bisphosphate	Uses ATP; committed regulated step
4	Aldolase: F1,6BP → GAP + DHAP	Six-carbon sugar splits into two trioses
5	Triose phosphate isomerase: DHAP → GAP	Two GAP now proceed per glucose
6	GAP dehydrogenase: GAP → 1,3-bisphosphoglycerate	2 NADH per glucose; uses inorganic phosphate
7	Phosphoglycerate kinase: 1,3-BPG → 3-phosphoglycerate	2 ATP per glucose
8	Phosphoglycerate mutase: 3-PG → 2-PG	Phosphate repositioning
9	Enolase: 2-PG → phosphoenolpyruvate	Water released
10	Pyruvate kinase: PEP → pyruvate	2 ATP per glucose; effectively irreversible

Net, using a common biochemical convention:

$$Glucose + 2ADP + 2P_i + 2NAD^+ \rightarrow 2Pyruvate + 2ATP + 2NADH + 2H^+ + 2H_2O$$

Net ATP = **2**, from 4 produced minus 2 consumed. NADH = **2**. No CO_2 is released during glycolysis. Steps 7 and 10 use **substrate-level phosphorylation**, unlike ATP synthase's gradient-driven ATP production.

Major control points

- **Hexokinase:** high-affinity glucose phosphorylation in many tissues; inhibited by G6P in common isoforms. **Glucokinase** in liver and pancreatic β cells has distinct kinetic/regulatory properties and is not inhibited by G6P in the same way.

- **PFK-1:** inhibited by high ATP and citrate, activated by AMP and fructose-2,6-bisphosphate; low pH can inhibit muscle glycolysis. ATP can be both substrate and allosteric inhibitor at different sites.
- **Pyruvate kinase:** activated feed-forward by F1,6BP; inhibited by ATP and alanine in common regulatory descriptions. In liver, glucagon/PKA signaling inhibits the L isoform through phosphorylation. Do not generalize this exact hormonal control to every tissue isoform.

Fructose-1,6-bisphosphate is a glycolytic intermediate. **Fructose-2,6-bisphosphate** is a powerful regulator; the names are not interchangeable.

Fermentation regenerates NAD^+

Lactate dehydrogenase reduces pyruvate to lactate while oxidizing NADH to NAD^+. This sustains GAP dehydrogenase and glycolysis. The fermentation reaction itself adds no ATP; glucose → 2 lactate gives net **2 ATP** from glycolysis. Mature erythrocytes depend on glycolysis because they lack mitochondria. Lactate production can also occur in oxygenated cells; "lactate always proves absence of oxygen" is too strong.

Yeast alcoholic fermentation uses pyruvate decarboxylase to make acetaldehyde + CO_2, then alcohol dehydrogenase to make ethanol while regenerating NAD^+. It also yields net 2 glycolytic ATP per glucose. Humans do not use alcoholic fermentation as a normal route for glucose metabolism.

Pyruvate dehydrogenase: the irreversible bridge

In the mitochondrial matrix, PDH converts pyruvate + CoA + NAD^+ → acetyl-CoA + CO_2 + NADH. For one glucose, double those products. The complex uses **TPP (B1), lipoamide, CoA (B5), FAD (B2), and NAD^+ (B3)**. Its oxidative decarboxylation is effectively irreversible in humans; acetyl-CoA cannot simply be converted back to pyruvate for net glucose synthesis.

High NADH and acetyl-CoA favor inhibition. PDH kinase phosphorylates/inactivates PDH; PDH phosphatase reverses it. Energy-demand signals such as Ca^{2+} in working muscle favor activation through tissue-dependent regulation. Distinguish this phosphorylation switch from the many enzymes activated by phosphorylation.

Citric acid cycle: carbon and electron accounting

The TCA cycle occurs mainly in the matrix; **succinate dehydrogenase is embedded in the inner membrane as complex II**.

Step	Conversion	Important product
Citrate synthase	Oxaloacetate (4C) + acetyl-CoA (2C) → citrate (6C)	CoA released
Aconitase	Citrate → isocitrate	Isomerization
Isocitrate dehydrogenase	Isocitrate → α-ketoglutarate (5C)	NADH + CO_2
α-Ketoglutarate dehydrogenase	α-KG → succinyl-CoA (4C)	NADH + CO_2
Succinyl-CoA synthetase	Succinyl-CoA → succinate	GTP or ATP equivalent
Succinate dehydrogenase	Succinate → fumarate	FAD-linked electron transfer to Q
Fumarase	Fumarate → malate	Hydration
Malate dehydrogenase	Malate → oxaloacetate	NADH

Per acetyl-CoA: **3 NADH + 1 $FADH_2$ equivalent + 1 GTP/ATP + 2 CO_2**. Oxaloacetate is regenerated, not net produced. Two CO_2 are released in each turn, but the entering acetyl carbons are not necessarily the ones released in the first turn. Carbon-labeling problems require tracking atoms rather than assuming immediate acetyl-carbon loss.

Citrate synthase, isocitrate dehydrogenase, and α-KG dehydrogenase are important control points. High NADH/energy abundance generally slows oxidative flux; ADP and Ca^{2+} can promote demand-responsive activity at relevant enzymes. Product removal helps drive unfavorable individual steps such as malate oxidation.

The cycle is **amphibolic**: oxidation supplies energy, while intermediates support biosynthesis. Withdrawal requires **anaplerosis**, especially pyruvate carboxylase replenishing oxaloacetate. Although no TCA step directly consumes O_2, sustained oxidative flux generally depends on the respiratory chain reoxidizing NADH/FAD-linked carriers.

Worked example 8: energy ledger for one glucose

Source	Direct ATP equivalents	NADH	$FADH_2$ equivalents
Glycolysis	2	2, cytosolic	0
Two PDH reactions	0	2	0
Two TCA turns	2	6	2
Total before respiratory conversion	4	10	2

Using about 2.5 ATP per matrix NADH and 1.5 per Q-level FAD-linked pair, total yield is about **32 ATP** with malate–aspartate transfer of glycolytic reducing equivalents, or **30 ATP** with the glycerol-3-phosphate shuttle. These are conventional estimates; leak, transport, and cellular conditions affect actual yield. If an exam supplies older 3/2 ratios, apply the specified convention and identify it.

Check yourself: Why does oxygen deprivation inhibit PDH/TCA even though neither pathway uses O_2 as its direct substrate? Which glycolytic step directly needs NAD^+? Why does fermentation maintain glycolysis without producing additional respiratory ATP?

10. Gluconeogenesis and reciprocal control

You should be able to: name the bypasses, calculate energy cost, identify usable carbon sources, and explain fasting regulation. **Sources:** [O], topic 10; [F, p. 47] mentions pyruvate carboxylation. Most detail is standard-knowledge expansion.

Gluconeogenesis synthesizes glucose primarily in the **liver**, with an important **renal cortex** contribution during prolonged fasting. Reactions span mitochondrial matrix, cytosol, and endoplasmic reticulum. It is not a simple reversal of glycolysis because the three effectively irreversible glycolytic steps need alternative reactions.

The bypass reactions

Glycolytic obstacle	Gluconeogenic bypass	Location/cofactor/energy
Pyruvate kinase	Pyruvate carboxylase: pyruvate → oxaloacetate	Matrix; biotin; ATP; activated by acetyl-CoA
Same obstacle, continued	PEP carboxykinase: OAA → PEP + CO_2	Uses GTP; cytosolic and/or mitochondrial isoforms
PFK-1	Fructose-1,6-bisphosphatase: F1,6BP → F6P	Hydrolysis; inhibited by AMP and F2,6BP
Hexokinase/glucokinase	Glucose-6-phosphatase: G6P → glucose	ER system; present in glucose-exporting tissues, absent in skeletal muscle

Oxaloacetate does not freely cross the inner mitochondrial membrane. Carbon can exit as malate/aspartate, or PEP can be generated in mitochondria and transported, depending on substrate and isoform context. The malate route also helps provide cytosolic reducing power. Do not draw OAA freely diffusing through the inner membrane.

From **two pyruvate to one glucose**, the pathway consumes **4 ATP + 2 GTP + 2 NADH**, or six high-energy phosphate equivalents plus reducing power. Two ATP are used by pyruvate carboxylase, two GTP by PEPCK, and two ATP in the reverse phosphoglycerate-kinase reaction. Starting from lactate supplies NADH via lactate dehydrogenase; this does not make the overall synthesis energetically free.

Carbon sources and restrictions

- **Lactate:** liver converts lactate to pyruvate and glucose; the Cori cycle shifts an energy burden to gluconeogenic organs.
- **Alanine and other glucogenic amino acids:** carbon skeletons enter as pyruvate or TCA intermediates after nitrogen handling.
- **Glycerol:** from triacylglycerol enters through glycerol-3-phosphate and DHAP in tissues with the required enzymes.
- **Odd-chain fatty acids:** propionyl-CoA can contribute net gluconeogenic carbon via succinyl-CoA.
- **Even-chain fatty acids:** acetyl-CoA supplies energy and regulatory signals but does **not** give net glucose carbon in humans. PDH is irreversible and the TCA cycle gives no net OAA gain from acetyl-CoA. The glycerol component of triacylglycerol is a separate exception to the broad statement that "fat cannot make glucose."

Reciprocal control prevents futile cycling

In liver, **AMP and F2,6BP activate PFK-1 and inhibit FBPase-1**, favoring glycolysis. Glucagon → cAMP → PKA phosphorylates the **liver PFK-2/FBPase-2 isoform**, lowering F2,6BP and favoring gluconeogenesis. Insulin generally promotes the opposite liver switch. This direction is isoform-specific; cardiac PFK-2 regulation differs.

Acetyl-CoA from fatty acid oxidation activates pyruvate carboxylase and signals that pyruvate can be conserved for glucose synthesis. High hepatic ATP/NADH from fatty acid oxidation supports the energy demand of gluconeogenesis. "Fasting" does not mean ATP must be low in every liver compartment.

Worked example 9: Cori-cycle cost. Peripheral anaerobic glycolysis yields 2 ATP per glucose converted to two lactate. Regenerating that glucose from two lactate consumes 6 ATP/GTP equivalents in liver. Net whole-cycle cost = **4 high-energy phosphate equivalents**. The benefit is sustaining peripheral ATP production and recycling carbon, not generating net whole-body ATP.

Check yourself: Why can muscle glycogen support contraction but not directly maintain blood glucose? How does F2,6BP coordinate two opposing pathways? Why are biotin and GTP needed in the pyruvate-to-PEP bypass?

11. Oxidative phosphorylation

You should be able to: trace electrons and protons separately, explain coupling, calculate ATP estimates, and distinguish inhibitors from uncouplers. **Sources:** [O], topic 11; [F, pp. 51–53]. Mechanistic detail and diagnostic patterns are expanded here.

Electron flow and proton pumping

The respiratory chain is in the **inner mitochondrial membrane**. Protons are moved from matrix to intermembrane space. Oxygen is reduced to water at complex IV; ATP synthase permits protons to return toward the matrix while producing ATP.

Component	Electron route	Approximate H^+ moved per electron pair
Complex I	NADH → FMN/Fe–S → ubiquinone Q	4
Complex II	Succinate → enzyme-bound FAD/Fe–S → Q	0
Q pool	Mobile lipid-soluble carrier → complex III	Included in complex accounting
Complex III	QH_2 → cytochrome c through Q cycle	4
Cytochrome c	Mobile one-electron carrier → complex IV	—
Complex IV	Electrons + O_2 + matrix H^+ → water	2 pumped; additional chemical H^+ used in water formation

Per NADH electron pair entering at I, about **10 H^+** are translocated; for entry at Q downstream of I, about **6 H^+**. Cytosolic NADH does not itself freely cross the inner membrane; shuttles transfer its reducing equivalents. Fatty-acid-oxidation electrons pass through ETF and ETF:Q oxidoreductase to Q, **not through complex II**. "$FADH_2$ enters at complex II" is only an appropriate shorthand for the succinate-dehydrogenase route, not all FAD-linked metabolism.

ATP synthase's F_0 portion conducts protons and drives rotation; F_1 catalytic sites interconvert binding states. The effective H^+/ATP accounting includes the synthase's stoichiometry and transport costs. Standard estimates of **2.5 ATP/NADH and 1.5 ATP per Q-level pair** are useful approximations, not exact universal molecular constants.

The matrix is normally more negative and more alkaline than the intermembrane space. Both electrical and chemical gradients contribute to proton-motive force. For an explicitly defined direction use the electrochemical free-energy equation from section 8; avoid memorizing an unsigned voltage expression without defining the sides.

Respiratory control

With substrate, oxygen, phosphate, and intact mitochondria, adding ADP promotes ATP synthesis and oxygen consumption. If ADP becomes limiting, the gradient builds, creating backpressure on electron transport. Energy demand raises ADP availability and promotes coupled respiration. Adenine nucleotide translocase exchanges matrix ATP for cytosolic ADP; phosphate transport also supports ATP synthesis.

Perturbation table

Assume adequate electron-donor supply and oxygen unless the perturbation removes them. Arrows describe the initial/typical steady response; exact behavior depends on substrate entry and experimental conditions.

Perturbation	O_2 consumption	Proton gradient	Oxidative ATP synthesis	Why
Rotenone, with NADH-linked substrates	Falls	Falls as dissipated	Falls	Complex I blocked; succinate can bypass I experimentally
Antimycin A	Falls	Falls as dissipated	Falls	Complex III blocked
Cyanide/CO at complex IV	Falls sharply	Falls as dissipated	Falls	Terminal electron transfer impaired
Oligomycin	Falls in tightly coupled mitochondria	Initially rises/remains high	Falls	ATP synthase proton channel blocked
Protonophore uncoupler	Often rises if donor/O_2 available	Falls	Falls	Proton leak bypasses ATP synthase; heat rises
Oxygen removal	Stops oxidative O_2 use	Dissipates	Stops	No terminal acceptor

An uncoupler does not rescue ATP synthesis after oligomycin. It can relieve proton backpressure and restore electron flow/oxygen consumption while ATP synthase remains blocked. With a complex IV block, an uncoupler cannot restore the blocked terminal electron transfer.

UCP1 provides regulated proton conductance in brown adipose tissue for thermogenesis. Not all uncoupling is pathological, but uncoupled oxidation captures less energy as ATP. Distinguish this physiological mechanism from claims that uncontrolled uncoupling is beneficial.

Worked example 10: 6 matrix NADH and 2 Q-level reducing equivalents give approximately $6(2.5) + 2(1.5) = 18$ ATP from oxidative phosphorylation. Add substrate-level ATP separately only if it belongs to the specified reaction basis.

Check yourself: Why does complex II not contribute the same ATP yield as complex I entry? Why can oligomycin and an uncoupler both lower ATP while having opposite effects on oxygen

consumption? Trace an electron from fatty acid oxidation to O_2 without incorrectly assigning its entry to complex II.

12. Fatty acid oxidation

You should be able to: explain activation/import, reconstruct the four reactions, calculate even-chain yields, and connect fasting to ketone production. **Sources:** [O], topic 12; [F, pp. 54–55]. Quantitative accounting and special cases are expanded here.

Activation and mitochondrial entry

Acyl-CoA synthetase activates fatty acid using ATP → AMP + PP_i; PP_i hydrolysis drives the reaction. This costs **two ATP equivalents**, not one, even though one ATP molecule is consumed. Location depends on enzyme and substrate; long-chain fatty-acid activation is associated with membranes including the outer mitochondrial membrane.

Long-chain acyl groups enter the matrix using the **carnitine shuttle**: CPT-I forms acylcarnitine on the outer-membrane side; the inner-membrane carnitine/acylcarnitine translocase exchanges substrates; CPT-II regenerates matrix acyl-CoA. The acyl group is transferred; long-chain acyl-CoA does not simply diffuse across the inner membrane.

Malonyl-CoA inhibits CPT-I, coordinating synthesis and oxidation. When fatty acid synthesis is favored, this reduces simultaneous mitochondrial entry for oxidation. AMPK inhibits acetyl-CoA carboxylase, lowering malonyl-CoA and favoring fatty acid entry/oxidation in relevant tissues.

Four recurring β-oxidation reactions

Order	Reaction	Product or cofactor
1	Acyl-CoA dehydrogenase: oxidation	trans-Δ²-enoyl-CoA; FAD-linked electrons to ETF → Q
2	Enoyl-CoA hydratase: hydration	L-3-hydroxyacyl-CoA
3	3-Hydroxyacyl-CoA dehydrogenase: oxidation	3-ketoacyl-CoA + NADH
4	β-Ketothiolase: thiolysis with CoA	Acetyl-CoA + acyl-CoA shortened by two carbons

For a saturated even-chain C_n fatty acid, cycles = $n/2-1$ and acetyl-CoA molecules = $n/2$. Each cycle yields one NADH and one Q-level FAD-linked pair. **The final four-carbon cleavage gives two acetyl-CoA**, so "one acetyl-CoA each cycle" needs this final-cycle qualification.

$$ATP_{net} = 10\left(\frac{n}{2}\right) + 4\left(\frac{n}{2} - 1\right) - 2 = 7n - 6$$

This shortcut assumes a **free, saturated, even-chain fatty acid**, complete oxidation through TCA/respiration, 2.5/1.5 ATP estimates, and no additional losses. It is not valid unchanged for unsaturated, odd-chain, or peroxisomal substrates. Starting with already activated acyl-CoA removes the activation deduction.

Worked example 11: palmitate, C16. Seven cycles produce 7 NADH, 7 Q-level pairs, and 8 acetyl-CoA. β-oxidation reducing equivalents yield $7(2.5)+7(1.5) = 28$ ATP; acetyl-CoA oxidation gives $8(10) = 80$; activation costs 2. Net = **106 ATP equivalents**. For palmitoyl-CoA supplied already activated, the corresponding estimate is 108.

Special substrates and fasting

- **Unsaturated fatty acids:** require isomerase and sometimes NADPH-dependent reductase; pre-existing double bonds can bypass a FAD-linked oxidation and reduce ATP yield relative to a comparable saturated chain.

- **Odd-chain fatty acids:** end in propionyl-CoA. Propionyl-CoA carboxylase requires biotin/ATP; methylmalonyl-CoA mutase requires B12 and forms succinyl-CoA.
- **Very-long-chain fatty acids:** often begin oxidation in peroxisomes; the first oxidation transfers electrons to oxygen, producing H_2O_2 handled by catalase, rather than conserving that step's electrons through mitochondrial ETF.
- **Fasting liver:** β-oxidation supplies NADH, ATP, and acetyl-CoA. As OAA is directed toward gluconeogenesis and its availability is constrained, acetyl-CoA increasingly supports ketogenesis. High NADH also affects TCA flux.

Ketone bodies are **acetoacetate, β-hydroxybutyrate, and acetone**; β-hydroxybutyrate is grouped physiologically with ketones although chemically it is not a ketone. Liver exports them but lacks the key SCOT/thiophorase reaction for their utilization. Extrahepatic mitochondria can oxidize them; red cells cannot because they lack mitochondria. During prolonged fasting, brain use of ketones reduces but does not eliminate glucose demand.

Check yourself: Why do activation costs equal two high-energy phosphate equivalents? Why does malonyl-CoA regulate CPT-I? Why can fasting liver make ketones while needing fatty acid oxidation to support glucose production?

13. Enzyme catalysis, kinetics, and inhibition

You should be able to: use Michaelis–Menten equations, interpret kinetic parameters, identify inhibitor patterns, and separate catalysis from thermodynamics. **Sources:** [O], topic 13; [F, pp. 18–26]. Derivations and numerical work are expanded here.

Catalytic mechanisms and model assumptions

Enzymes stabilize the transition state relative to the ground state using orientation/proximity, acid–base catalysis, transient covalent chemistry, or metal-ion catalysis. They accelerate forward and reverse reactions without changing the reaction's equilibrium constant. Tight binding of substrate alone is not a sufficient explanation for catalysis; excessive ground-state stabilization can impede reaction.

For $E + S \rightleftharpoons ES \rightarrow E + P$, the simple Michaelis–Menten model uses initial rates, negligible product accumulation/reverse reaction, substrate in excess of enzyme, and a quasi-steady ES concentration after a short transient. Many multisubstrate, cooperative, or complex mechanisms require more elaborate models.

$$v_0 = \frac{V_{max}[S]}{K_m + [S]}, \qquad V_{max} = k_{cat}[E]_T, \qquad K_m = \frac{k_{-1} + k_{cat}}{k_1}$$

K_m has concentration units and gives the substrate concentration at half-maximal initial velocity under this model. It equals the ES dissociation constant only in an appropriate limiting regime, such as $k_{cat} \ll k_{-1}$. **A smaller K_m is not universally proof of tighter physical binding.** k_{cat} has time^{-1} units; k_{cat}/K_m has concentration^{-1} time^{-1} units and measures low-substrate catalytic efficiency in this model.

At $[S] \ll K_m$, $v \approx (k_{cat}/K_m)[E]_T[S]$. At $[S] = K_m$, $v = V_{max}/2$. At $[S] \gg K_m$, $v \approx V_{max}$. Doubling active enzyme doubles V_{max}, not the intrinsic K_m, provided the model and conditions remain appropriate.

Worked example 12: velocity and turnover

If $V_{max} = 120$ µmol min^{-1} and $K_m = 3.0$ mM, at $[S] = 6.0$ mM the rate is $120(6)/(3+6) = 80$ µmol min^{-1}. If total active enzyme is 0.20 µmol, $k_{cat} = 120/0.20 = 600$ min^{-1} $= 10$ s^{-1}. Here V_{max} is amount/time and enzyme is amount; if V_{max} were concentration/time, use enzyme concentration instead.

Reversible inhibitors

Define $\alpha = 1 + [I]/K_i$ for inhibitor binding free E, and $\alpha' = 1 + [I]/K_i'$ for binding ES. A common mixed-inhibition model gives:

$$v = \frac{V_{max}[S]}{\alpha K_m + \alpha'[S]}, \quad V_{max,app} = \frac{V_{max}}{\alpha'}, \quad K_{m,app} = \frac{\alpha}{\alpha'}K_m$$

Pattern	Binding in this model	$K_{m,app}$	$V_{max,app}$	Reciprocal-plot pattern
Competitive	E, not ES	Increases	Same	Same y-intercept
Uncompetitive	ES, not E	Decreases	Decreases by same factor	Parallel lines
Pure noncompetitive	E and ES equally	Same	Decreases	Same x-intercept
Mixed	E and ES unequally	Up if α > α′; down if α < α′	Decreases	General intersection off the axes

Competitive inhibition can be overcome in the limit of sufficiently high substrate in the simple reversible model. Pure noncompetitive inhibition is a **special case** of mixed inhibition. Binding at a distinct/allosteric site alone does not establish a pure noncompetitive kinetic pattern. Competitive kinetics also does not require an oversimplified claim that every inhibitor must physically occupy the exact substrate site.

Irreversible inhibition causes persistent inactivation, often through covalent modification. More substrate added after inactivation does not restore lost active enzyme. Time dependence, dilution/dialysis behavior, and direct biochemical evidence help distinguish irreversible inhibition from reversible inhibition. Some tight-binding inhibitors are reversible but dissociate slowly; one observation alone may not establish mechanism.

Worked example 13: identifying and quantifying competition

An enzyme has $V_{max} = 100$ µM/min and $K_m = 2$ mM. With 3 mM inhibitor, V_{max} remains 100 and apparent $K_m = 8$ mM. Thus α = 8/2 = 4, and $K_i = [I]/(\alpha - 1) = 3/3 = 1$ mM. At substrate = 2 mM, uninhibited rate = 50 µM/min and inhibited rate = $100(2)/(8 + 2) = 20$ µM/min. This dataset is consistent with competitive inhibition under the assumed model.

Reading data and allostery

$$\frac{1}{v} = \frac{K_m}{V_{max}}\frac{1}{[S]} + \frac{1}{V_{max}}$$

Lineweaver–Burk intercepts are $1/V_{max}$ and $-1/K_m$. Reciprocal transformations magnify errors at low substrate; nonlinear fitting of the original rates is generally preferable for parameter estimation. For exam plots, identify axes before comparing slopes/intercepts.

Allosteric enzymes may show sigmoidal kinetics and use $K_{0.5}$ rather than a simple Michaelis–Menten K_m. Allostery need not always produce cooperativity. Kinases transfer phosphate, usually from ATP; phosphatases hydrolyze phosphate esters. **Phosphorylation can activate or inhibit** depending on the protein and context.

Check yourself: Why is K_m not automatically K_d? Does an unchanged K_m with lower V_{max} prove a specific molecular binding site? Why is enzyme amount necessary to calculate k_{cat}?

14. Hormone signaling and metabolic integration

You should be able to: compare insulin, glucagon, and epinephrine; follow fed/fasted changes; and explain tissue specificity. **Sources:** [O], topic 14; [F, pp. 24, 48, 57] for partial supporting concepts. Most hormonal mechanisms are standard-knowledge expansion.

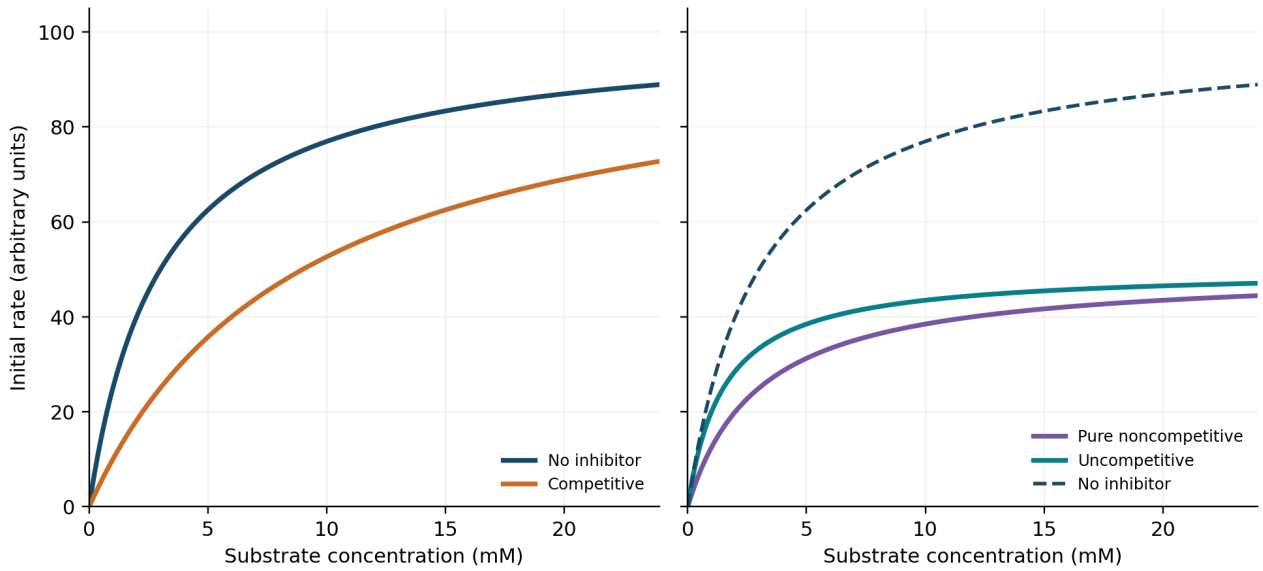


Figure 2: Model initial-rate curves illustrating different apparent kinetic parameters. The uncompetitive and pure noncompetitive examples both halve Vmax but differ in their effects on apparent Km.

Major hormonal logic

Hormone	Receptor/major route	Main metabolic context
Insulin	Receptor tyrosine kinase → IRS → PI3K → PIP_3 → Akt	Fed state; uptake/storage and anabolic signaling
Glucagon	Predominantly hepatic Gs-coupled receptor → cAMP → PKA	Fasting; hepatic glucose output
Epinephrine	β-adrenergic Gs and α_1-adrenergic Gq routes, depending on tissue	Acute stress; fuel mobilization
Cortisol	Intracellular/nuclear receptor	Longer-term transcriptional adaptation; permissive/catabolic effects

Insulin promotes **GLUT4 translocation in skeletal muscle and adipose tissue**. Do not generalize GLUT4 control to all glucose uptake: liver commonly uses GLUT2 and erythrocytes GLUT1. Hepatic insulin action includes enzyme regulation and gene-expression changes, not insulin-driven GLUT4 insertion.

Glucagon promotes liver glycogenolysis and gluconeogenesis. **Skeletal muscle is not a major direct glucagon target**; its glycogen is principally used locally, with epinephrine, Ca^{2+}, and energy signals important for mobilization. Human adipose lipolysis is strongly influenced by catecholamines and insulin; avoid assuming that all species' direct glucagon effects on adipocytes are identical.

Glycogen switching

In liver, cAMP/PKA activates phosphorylase kinase, which activates glycogen phosphorylase; PKA-related phosphorylation also inhibits glycogen synthase. Insulin favors storage through coordinated Akt and phosphatase signaling, including inhibition of GSK3 and promotion of glycogen-synthase activation. Glycogen phosphorylase releases mostly glucose-1-phosphate from α(1→4) bonds, while debranching handles α(1→6) junctions. Liver G6Pase permits free glucose export; muscle lacks that export step.

One hormone, coordinated targets: glucagon can stimulate liver glycogen breakdown, lower F2,6BP to favor gluconeogenesis, and inhibit liver pyruvate kinase. These effects preserve/export glucose rather than immediately consuming it through hepatic glycolysis.

Tissue	Fed-state emphasis	Fasting emphasis
Liver	Glycogen storage, glycolysis, lipid synthesis	Glycogenolysis initially; gluconeogenesis, β-oxidation, ketogenesis increasingly
Muscle	Insulin-stimulated uptake; glycogen/protein synthesis	Greater fatty acid use at rest; local glycogen use with work
Adipose	Glucose uptake and triacylglycerol storage; lipolysis suppressed	Lipolysis supplies fatty acids and glycerol as hormonal conditions favor it
Brain	Primarily glucose	Increasing ketone use in prolonged fasting; some glucose still required
Mature RBCs	Glycolysis → lactate	Still glycolysis → lactate; cannot use fatty acids/ketones oxidatively

Acute signaling operates in seconds/minutes through existing proteins; transcriptional responses often take longer. Hormone concentration, receptor distribution, isoforms, substrate availability, and cellular energy state all shape the response.

Case reasoning: after an overnight fast, lower insulin and higher glucagon favor liver glucose output. Adipose-derived fatty acids support hepatic energy production; glycerol contributes glucose carbon. RBCs continue releasing lactate, which can return to liver. During prolonged fasting, ketone supply reduces brain glucose demand and protein breakdown pressure.

Check yourself: Why can the same phosphorylation cascade activate glycogen breakdown while inhibiting synthesis? Why does insulin increase muscle glucose uptake without the same GLUT4 mechanism in liver? Why must a hormone answer specify the tissue?

15. Caloric restriction and aging

You should be able to: explain proposed nutrient-sensing mechanisms and evaluate evidence without treating a model-organism result as proof of human lifespan extension. **Sources:** [O], topic 15. No dedicated coverage was found in [F]; this section is standard-knowledge expansion, not a verified summary of the uploaded textbook.

Caloric restriction (CR) means reduced energy intake without malnutrition. It is not synonymous with starvation, a particular fasting schedule, or restricting a single macronutrient. Nutrient composition, timing, baseline intake, genetics, age, sex, and environment can all modify responses.

Mechanistic framework

Sensor/pathway	General signal	Mechanistic link to maintenance
AMPK	Cellular energy stress, reflected in AMP/ADP relative to ATP	Promotes energy-generating pathways; restrains energy-intensive synthesis; supports autophagy regulation
Insulin/IGF signaling	Nutrient/growth-factor availability	PI3K/Akt affects growth, metabolism, and FOXO transcription factors
mTORC1	Integrated amino-acid, growth-factor, and energy status	Promotes growth/protein synthesis; suppresses autophagy when active
Sirtuins	NAD^+-dependent protein deacylation and metabolic context	Regulate stress/metabolic programs; relationships are tissue- and context-dependent
Autophagy/mitophagy	Nutrient/stress signals and organelle quality	Recycling and removal of damaged cellular components

A plausible integrated model is: reduced nutrient/growth signaling and appropriate energy sensing → less mTORC1-driven anabolism and altered AMPK/FOXO-related programs → greater maintenance, stress resistance, and cellular recycling. This is a network, not a universal single linear switch. Tissue energy status under CR is not simply "ATP depleted everywhere."

Mitochondrial adaptation, altered inflammation, proteostasis, and oxidative-stress responses may contribute. Reactive oxygen species are both damaging agents and signaling molecules; the claim "CR works only by reducing all ROS" is inadequate. Likewise, increasing NAD^+ or activating one sirtuin does not automatically reproduce every effect of CR.

Evidence and experimental interpretation

CR extends lifespan in many experimental models under particular conditions, but effects vary. Human studies can examine weight, cardiometabolic markers, functional outcomes, adherence, and adverse effects; these do **not by themselves demonstrate extension of maximum human lifespan**. Distinguish **lifespan**, **healthspan**, and surrogate biomarkers.

For an exam research question, propose an ad-libitum control, nutritionally adequate CR group, controlled diet composition, monitored intake/body mass, and relevant sex/age/genotype strata. If studying a drug mimic, include appropriate vehicle and intake-matched controls where needed to separate a direct drug effect from reduced food consumption. Measure pathway engagement as well as functional outcomes. Correlation between AMPK activation and survival does not establish necessity; loss-of-function/rescue designs can test causation, subject to their own confounders.

Model short-answer paragraph: Caloric restriction may shift resource allocation from growth toward maintenance through nutrient-sensing networks involving insulin/IGF signaling, AMPK, mTORC1, and NAD^+-dependent enzymes. Reduced mTORC1 activity can permit autophagy, while stress-response programs support proteostasis and organelle quality. The magnitude and direction of effects depend on organism and context. Improvements in metabolic biomarkers support a physiological effect but do not establish longer human lifespan.

Check yourself: Why must CR avoid malnutrition in its definition? What experiment distinguishes a correlated signaling marker from a necessary mediator? Why is a biomarker change weaker evidence than a directly measured survival outcome?

16. G-protein signaling

You should be able to: trace activation and termination, compare Gs/Gi/Gq, and predict persistent-signaling mutations. **Sources:** [O], topic 16. This section is standard-knowledge expansion; [F] contains no dedicated G-protein pathway section.

The heterotrimeric G-protein cycle

A typical GPCR has **seven transmembrane helices**. In the inactive state, Gα carries GDP and associates with Gβγ. Ligand-bound receptor acts as a guanine-nucleotide exchange factor: GDP leaves and GTP binds Gα. The receptor does **not** phosphorylate GDP into GTP. Gα-GTP and Gβγ can then regulate effectors through rearrangement and/or separation of the signaling complex.

Gα's intrinsic GTPase hydrolyzes GTP to GDP, terminating its active state; RGS proteins accelerate this process for many Gα families. Reassociation restores the inactive heterotrimer. Gβγ is also a signaling unit, not merely an inert anchor.

Gα family	Main effector route	Typical downstream effect
Gs	Stimulates adenylyl cyclase	cAMP rises → PKA activation
Gi	Inhibits many adenylyl cyclase isoforms	cAMP tends to fall; Gβγ can regulate channels/other targets
Gq	Activates PLCβ	$PIP_2 \rightarrow IP_3$ + DAG; Ca^{2+} signaling and PKC activation

cAMP pathway

Ligand → GPCR → Gs-GTP → adenylyl cyclase converts ATP to cAMP → cAMP binds PKA regulatory subunits → catalytic activity changes target phosphorylation. Phosphodiesterases convert cAMP to AMP, phosphatases reverse target phosphorylation, and receptor desensitization limits further stimulation.

IP_3/DAG pathway

Ligand → Gq → PLCβ cleaves membrane PIP_2. **IP_3 is soluble** and opens IP_3-receptor Ca^{2+} channels on ER; **DAG remains in the membrane**. DAG and, for conventional PKC isoforms, Ca^{2+} promote PKC activation. Ca^{2+} also binds calmodulin and regulates other enzymes. Pumps/sequestration, IP_3 metabolism, and DAG metabolism help terminate the response.

Amplification and termination

One active receptor can activate multiple G proteins, an enzyme can generate many second messengers, and downstream kinases can modify many targets. Amplification coexists with spatial localization and negative feedback. GPCR kinases phosphorylate activated receptors; β-arrestin binding can reduce G-protein coupling and promote internalization. Internalization may lead to recycling, degradation, or other signaling outcomes.

Perturbation examples: a Gα mutant unable to hydrolyze GTP can prolong signaling downstream even after ligand removal. Cholera toxin maintains stimulatory Gαs signaling through impaired GTPase shutoff, increasing cAMP in affected cells. Pertussis toxin prevents normal receptor-mediated activation of Gi, removing an inhibitory route in susceptible contexts. These are different lesions, not interchangeable examples of G-protein activation.

Check yourself: Which steps use nucleotide exchange versus hydrolysis? Why can phosphodiesterase inhibition increase cAMP without directly activating adenylyl cyclase? Which second messenger stays in the membrane, and which mobilizes ER Ca^{2+}?

17. Membrane receptor signaling and pathway comparison

You should be able to: distinguish receptor classes, follow insulin and growth-factor cascades, and identify where a perturbation acts. **Sources:** [O], topic 17. This section is standard-knowledge expansion; [F] does not provide dedicated receptor-signaling coverage.

Receptor class	Activation principle	Example and downstream route
GPCR	Ligand alters receptor-mediated heterotrimeric G-protein nucleotide exchange	Glucagon/β-adrenergic → cAMP
Receptor tyrosine kinase (RTK)	Ligand drives productive receptor rearrangement/association and tyrosine phosphorylation	Insulin receptor; growth-factor RTKs
Cytokine receptor with associated kinase	Receptor recruits/activates a separate kinase	JAK → STAT; receptor itself lacks intrinsic tyrosine-kinase activity
Ligand-gated ion channel	Ligand changes channel opening	Nicotinic acetylcholine receptor
Receptor guanylyl cyclase	Ligand increases cGMP formation	Natriuretic peptide receptor

Lipophilic steroid and thyroid hormones often act through intracellular/nuclear receptors and transcriptional regulation. Their receptors are a useful contrast but are not all membrane receptors. Receptor class predicts a mechanism, not a universal response speed or single downstream outcome.

RTKs and insulin

Many RTKs dimerize upon ligand binding, bringing kinase domains into an activation arrangement and generating phosphotyrosine docking sites. **The insulin receptor is already a**

disulfide-linked $\alpha_2\beta_2$ complex, so "all RTKs activate only by ligand-induced dimerization of separate monomers" is an overgeneralization.

Insulin → receptor tyrosine phosphorylation → IRS recruitment/phosphorylation → PI3K converts PIP_2 to **PIP_3** → recruitment and activation machinery for Akt → changes in GLUT4 trafficking, glycogen synthesis, and other metabolic processes. PDK1 and mTORC2 contribute to full Akt activation. PTEN dephosphorylates PIP_3 back toward PIP_2, providing a negative regulatory route.

Growth-factor RTKs can recruit adaptor/GEF machinery, for example Grb2/SOS, to activate **Ras**, a small monomeric GTPase. Ras-GTP → Raf → MEK → ERK links signals to gene expression and proliferation. Ras is **not** the heterotrimeric Gαβγ complex, although both use GDP/GTP switching. Ras-GAP stimulates GTP hydrolysis; GEFs promote exchange.

Some RTKs also activate **PLCγ**, leading to IP_3/DAG production. Contrast **Gq → PLCβ** with **RTK → PLCγ**. Likewise, **PI3K makes PIP_3**, while **PLC cleaves PIP_2**. Confusing these phospholipid enzymes produces incorrect downstream predictions.

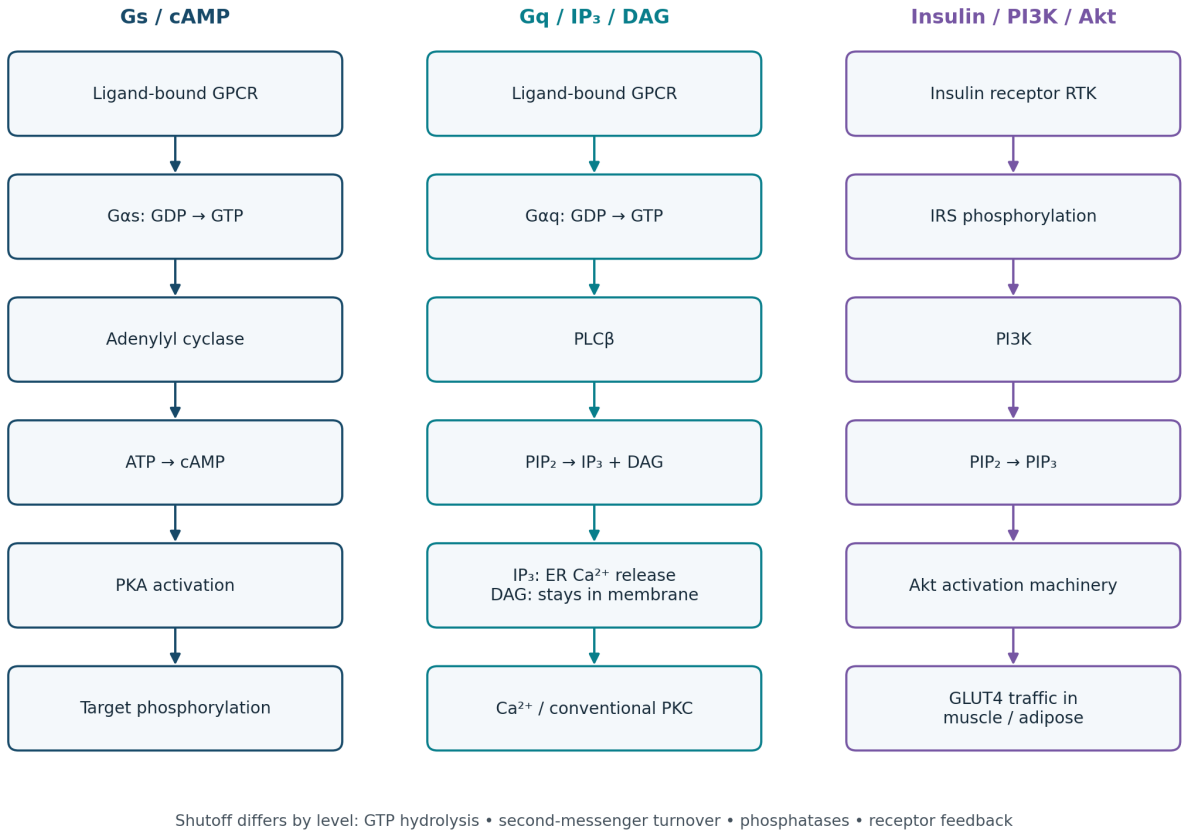


Figure 3: Three signaling routes compared. These are simplified core routes; the text explains tissue specificity, additional activation steps, and termination mechanisms.

Reading a pathway experiment

Suppose insulin produces normal receptor phosphorylation but no rise in Akt activation, and GLUT4 translocation fails. The defect could lie between receptor docking and Akt, not necessarily in insulin binding. Test IRS/PI3K/PIP_3 and Akt activation rather than naming one unmeasured culprit as proven. A downstream rescue can help locate a lesion, but interpretation depends on specificity and expression levels.

Signal specificity emerges from receptor distribution, scaffolds, subcellular localization, dose/duration, feedback, and phosphatases. Shared kinases do not make all ligand responses identical. A constitutively active downstream component may evade normal receptor shutoff.

Check yourself: Why is insulin signaling classified as RTK signaling rather than Gs signaling?

How does a JAK-associated receptor differ from an RTK? How can two receptors produce different outputs despite both affecting MAPK?

18. Integration tools and supplementary concepts

One-page mental map: where fuel carbon goes

Starting material or node	Main destinations	Constraint to remember
Glucose	Glycolysis; glycogen; biosynthetic routes	Entry and fate vary by tissue/hormones
Pyruvate	Lactate; acetyl-CoA by PDH; OAA by pyruvate carboxylase; alanine	PDH is irreversible; carboxylase requires ATP/biotin
Acetyl-CoA	TCA; ketones in liver; lipid/cholesterol synthesis through compartmental transfer routes	No net glucose from even-chain acetyl-CoA in humans
TCA intermediates	Oxidation and biosynthesis	Withdrawal requires anaplerotic replenishment
Fatty acids	β-oxidation → acetyl-CoA + reducing equivalents	Long-chain entry regulated at CPT-I
Glycerol	DHAP → glycolysis or gluconeogenesis	Distinct from fatty-acid carbon
NADH	Respiratory electron transfer or reductive reactions such as LDH	Cytosolic/mitochondrial pools need transfer mechanisms

Carbon rule: count carbons separately from ATP. **Electron rule:** track NADH and FAD-linked equivalents before converting them to ATP. **Compartment rule:** do not assume intermediates cross the inner mitochondrial membrane freely. **Regulation rule:** specify tissue and isoform when a switch is not universal.

Supplement: nitrogen handling

This material appears in [F, pp. 56–58] but is not a standalone topic in [O]. Know the conceptual link to amino-acid carbon sources; do not prioritize a detailed urea-cycle memorization over a listed topic without additional course guidance.

Transamination: an aminotransferase commonly transfers an amino group to α-ketoglutarate, forming glutamate and an α-keto acid; pyridoxal phosphate (B6) is the key cofactor. This step does not directly release free ammonia. **Oxidative deamination:** glutamate dehydrogenase can release ammonium and regenerate α-ketoglutarate. Nitrogen moves safely between tissues largely as glutamine or alanine.

Liver converts nitrogen to urea; kidneys excrete it. In the urea cycle the two nitrogen atoms derive from free ammonium and aspartate, and the carbonyl carbon derives from bicarbonate. The cycle spans mitochondria and cytosol and consumes 3 ATP molecules corresponding to 4 high-energy phosphate equivalents. These details are supplementary to the official list.

Supplement: laboratory and vocabulary checks

Proteomics studies the proteins expressed in a particular system under particular conditions; the proteome is dynamic. Histones are Lys/Arg-rich basic proteins that associate with negatively charged DNA. Peptide bonds connect amino acids, glycosidic bonds link sugars, ester bonds link glycerol and fatty acids in triacylglycerols, and phosphodiester bonds connect nucleotides in nucleic-acid backbones. Dehydrogenases perform redox reactions; kinases transfer phosphoryl groups; phosphatases remove phosphate through hydrolysis. See [F, pp. 42–44].

Common mistakes worth eliminating

1. Using original buffer amounts after strong-acid addition instead of neutralizing first.
2. Treating pI as absence of all charged groups rather than zero average net charge.
3. Counting internal peptide bonds as free amino/carboxyl termini.
4. Calling denaturation ordinary peptide-bond hydrolysis.
5. Confusing high Hb affinity with improved oxygen delivery.

6. Treating D/L as the sign of optical rotation, or every ketose as nonreducing.
7. Calling all protein-mediated transport active.
8. Forgetting the electrical term for ions or mixing volts and millivolts.
9. Counting glycolytic 4 ATP production as 4 ATP net.
10. Omitting PDH NADH from complete glucose oxidation.
11. Treating TCA as a net producer of OAA from acetyl-CoA.
12. Making gluconeogenesis a direct reversal of the three irreversible glycolytic steps.
13. Saying all FAD-linked electrons enter through complex II.
14. Claiming an uncoupler increases oxidative ATP yield because O_2 use rises.
15. Deducting only one ATP equivalent for fatty-acid activation.
16. Equating every allosteric inhibitor with pure noncompetitive inhibition.
17. Treating K_m as an unconditional binding constant.
18. Assuming phosphorylation always activates a protein.
19. Assigning direct glucagon control or insulin-driven GLUT4 uptake to every tissue.
20. Confusing PI3K, PLC, PIP_2, PIP_3, IP_3, and DAG.
21. Saying GPCRs phosphorylate GDP to GTP or confusing Ras with Gαβγ.
22. Concluding human lifespan extension from an aging-related biomarker alone.

19. Original practice paper

Suggested practice conditions: 120 minutes, closed notes, scientific calculator, and your formulas-only sheet. **24 questions, 100 practice marks.** This timing, marking, and distribution are created for revision; they are not an official exam format or prediction. Questions 22–23 use supplementary nitrogen/assay material from [F]. Attempt the paper before reading the answers.

Questions

1. Buffer perturbation — 4 marks. A buffer with $pK_a = 7.20$ contains 8.0 mmol HA and 12.0 mmol A^-. Calculate its initial pH and its pH after adding 3.0 mmol HCl. Assume the conjugate pair remains appropriate. Explain why dilution alone does not produce the same pH change.

2. Cellular free energy — 4 marks. At 310 K, a reaction has $\Delta G^{\circ\prime} = +3.0$ kJ/mol and $Q = 0.050$. Calculate ΔG using $R = 0.008314$ kJ mol^{-1} K^{-1}. Would adding an enzyme change this value or the equilibrium constant?

3. Amino-acid titration — 4 marks. Lysine has supplied pK_a values 2.2, 9.0, and 10.5. Sketch its sequence of net charges as pH rises and estimate its pI. State the approximate charge of a free lysine at pH 7.

4. Structure and separation — 4 marks. A native 100 kDa protein contains two identical chains. Nonreducing SDS-PAGE gives a 100 kDa band and reducing SDS-PAGE a 50 kDa band. Give a consistent structural explanation. Would ordinary heat denaturation necessarily cleave its peptide backbone?

5. Oxygen delivery — 4 marks. Local tissue pH falls while CO_2 rises. Predict the Hb oxygen curve shift, the direction of P_{50} change, and the effect on unloading at fixed tissue pO_2. Explain why myoglobin does not show the same cooperative mechanism.

6. Antibody binding — 4 marks. An antibody site has $K_d = 5$ nM. Calculate occupancy at 20 nM free antigen in an independent-site model. Distinguish affinity from avidity and identify the antigen-binding and main effector portions of IgG.

7. Sugar chemistry — 4 marks. Identify the anomeric carbon of glucose and fructose. Explain why maltose is reducing but sucrose is not. Does D-fructose necessarily rotate polarized light to the right?

8. Membrane composition — 4 marks. At fixed temperature, a membrane shifts toward longer, more saturated acyl chains. Predict the packing/fluidity trend. Explain why cholesterol's effect cannot be described as "always increases fluidity."

9. Electrochemical transport — 4 marks. A monovalent cation moves from outside (100 mM) to inside (10 mM) at 310 K. Inside is −50 mV relative to outside. Calculate the transfer ΔG with $R = 8.314$ J mol^{-1} K^{-1} and $F = 96485$ C/mol. Can the entry support uptake of a neutral solute against its concentration gradient?

10. Glycolysis and fermentation — 4 marks. State glycolytic net ATP, NADH, and pyruvate per glucose. If both pyruvate are reduced to lactate, what is the net NADH balance for glucose → lactate, and why is this route useful?

11. PDH and TCA accounting — 4 marks. Starting with two pyruvate, count NADH, Q-level FAD-linked equivalents, GTP/ATP, and CO_2 from PDH plus two TCA turns. Do not include glycolysis or convert reducing equivalents to ATP.

12. Gluconeogenesis — 5 marks. List the four bypass enzymes and explain the 4 ATP + 2 GTP cost from two pyruvate to one glucose. Explain why acetyl-CoA from even-chain fatty acids cannot provide net glucose carbon in humans.

13. Respiratory perturbations — 5 marks. Isolated, well-coupled mitochondria respire on NADH-linked substrates with oxygen and ADP available. Predict ATP synthesis, oxygen consumption, and gradient changes after oligomycin. Then add a protonophore. Explain which processes recover and which do not.

14. Fatty-acid energy calculation — 5 marks. Calculate β-oxidation cycles, acetyl-CoA yield, and net ATP equivalents for free saturated C18 fatty acid. Use 2.5 ATP/NADH, 1.5 ATP/Q-level pair, and 10 ATP/acetyl-CoA. Include activation cost.

15. Kinetic parameters — 5 marks. An enzyme has $V_{max} = 90$ µmol/min, $K_m = 3$ mM, and active enzyme amount 0.15 µmol. Calculate velocity at 6 mM substrate, k_{cat} in s^{-1}, and k_{cat}/K_m in $M^{-1}s^{-1}$. State one reason K_m may differ from binding K_d.

16. Inhibitor diagnosis — 4 marks. Baseline parameters are $V_{max} = 80$ µM/min and $K_m = 4$ mM. With 6 mM inhibitor, $V_{max} = 80$ and $K_{m,app} = 12$ mM. Identify the consistent pattern, calculate K_i in the simple model, and calculate inhibited velocity at 12 mM substrate.

17. Hormone and tissue specificity — 4 marks. Explain how insulin increases glucose uptake in muscle and why the same transporter explanation should not be applied to liver. Why does glucagon not directly mobilize muscle glycogen in the same principal way as liver glycogen?

18. G-protein shutoff — 4 marks. A Gαs mutant binds GTP but cannot hydrolyze it efficiently. Predict cAMP signaling after ligand removal. Name two other mechanisms that can reduce the overall pathway response and distinguish exchange from hydrolysis.

19. Receptor signaling — 4 marks. Contrast GPCR and insulin-receptor activation. Distinguish the products of PI3K and PLC, and identify the second messenger that releases ER Ca^{2+}.

20. Aging evidence — 4 marks. A study finds that a CR-like intervention lowers an aging-associated biomarker and increases AMPK phosphorylation in humans for six months. What can and cannot be concluded about lifespan? Describe one mechanistic experiment and one important control.

21. Integrated fasting case — 4 marks. Explain why liver fatty-acid oxidation, gluconeogenesis, and ketogenesis can all increase during fasting. Identify what fatty acids contribute to glucose production and what they generally do not contribute.

22. Supplement: nitrogen handling — 4 marks. Contrast transamination and oxidative deamination. Name two major nitrogen-transport molecules and identify the organ converting nitrogen to urea.

23. Supplement: immunoassay interpretation — 4 marks. In a sandwich ELISA, a high-concentration sample gives less signal than the same sample after dilution. Give a plausible explanation, a useful check, and a reason that signal should not be converted to concentration without a standard curve and validated range.

24. Mechanistic synthesis — 4 marks. A liver-targeted intervention inhibits PFK-2 activity and increases FBPase-2 activity. Predict F2,6BP, PFK-1, FBPase-1, and the glycolysis/gluconeogenesis balance. State why you should specify the isoform and tissue before extrapolating this answer elsewhere.

20. Worked answers and marking guide

Award credit for correct reasoning and units. The allocations below are revision aids, not official marking criteria. A numerically correct answer produced by an inconsistent convention should be corrected before moving on.

1. Initial pH = $7.20 + \log_{10}(12/8) = 7.376 \approx 7.38$ (1). HCl converts 3 mmol A^- to HA: new amounts 9 and 11 mmol (1). New pH = $7.20 + \log_{10}(9/11) = 7.113 \approx 7.11$ (1). Dilution approximately preserves the conjugate ratio but lowers capacity; acid addition changes the ratio (1).

2. $\Delta G = 3.0 + (0.008314)(310)\ln 0.050 = -4.72$ kJ/mol (2). Forward reaction is thermodynamically favorable under these conditions (1). Enzyme changes rate, not ΔG or K_{eq} (1).

3. Charge sequence as groups deprotonate: +2 → +1 → 0 → −1 (1). The neutral form lies between 9.0 and 10.5 (1); $pI = 9.75$ (1). At pH 7 free lysine is approximately +1 (1).

4. Two 50 kDa chains are linked by one or more interchain disulfides (2). SDS disrupts many noncovalent interactions but a reducing reagent is needed to break those covalent links (1). Ordinary denaturation need not hydrolyze peptide bonds; it changes higher-order structure (1). The data support this explanation rather than proving every detail of native arrangement.

5. Right shift (1), higher P_{50}/lower affinity (1), and greater unloading at fixed tissue oxygen pressure (1). Mb has one binding site and lacks Hb's multisubunit cooperative T/R transition (1). H^+/CO_2 favor lower-affinity Hb conformations.

6. $\theta = 20/(5+20) = 0.80$ (1). Affinity is a single interaction's strength; avidity includes combined multivalent effects (1). Fab/variable domains bind antigen (1); Fc contributes effector interactions (1).

7. Glucose C1 and fructose C2 are anomeric (1). Maltose retains a free anomeric center (1), whereas sucrose links both anomeric centers (1). D/L specifies configuration, not optical-rotation sign; D-fructose need not be dextrorotatory (1).

8. Longer saturated chains pack more tightly and generally reduce fluidity/raise transition temperature (2). Cholesterol restrains motion at higher temperatures but disrupts ordered packing and solidification at lower temperatures (2). Composition modifies these trends.

9. Chemical term $= (8.314)(310)\ln 0.1 = -5.93$ kJ/mol (1); electrical term $= (96485)(-0.050) = -4.82$ kJ/mol (1). Total = **−10.76 kJ/mol**, favoring entry (1). Coupled neutral-solute uptake is possible if its positive chemical work and other costs do not exceed the available favorable free energy for the actual stoichiometry (1).

10. Glycolysis gives 2 ATP net, 2 NADH, and 2 pyruvate (2). LDH consumes those 2 NADH, so overall glucose → 2 lactate gives no net NADH (1). It regenerates NAD^+ needed at GAP dehydrogenase, sustaining substrate-level ATP production (1).

11. Two PDH reactions yield 2 NADH and 2 CO_2. Two TCA turns yield 6 NADH, 2 FAD-linked equivalents, 2 GTP/ATP, and 4 CO_2. Combined: **8 NADH, 2 FAD-linked equivalents, 2 GTP/ATP, 6 CO_2** (1 each). Glycolytic NADH is correctly excluded.

12. Pyruvate carboxylase, PEPCK, FBPase-1, and G6Pase (2). Two ATP at pyruvate carboxylase, two GTP at PEPCK, and two ATP in reverse phosphoglycerate kinase explain the six high-energy phosphate equivalents (2). Even-chain acetyl-CoA cannot reverse PDH and gives no net OAA gain through human TCA; thus no net glucose carbon from that source (1). Two NADH are also required starting from pyruvate.

13. Oligomycin blocks ATP synthase proton flow: ATP synthesis falls sharply (1); gradient initially rises/remains high (1), inhibiting electron flow and lowering O_2 consumption through backpressure (1). A protonophore dissipates the gradient and can restore/increase respiratory electron flow with donor/O_2 present (1), but does not restore ATP synthesis through blocked ATP synthase; energy becomes heat (1).

14. C18 gives 8 cycles (1) and 9 acetyl-CoA (1). Cycle reducing equivalents = 8 NADH + 8 Q-level pairs → $8(2.5 + 1.5) = 32$ ATP (1). Acetyl-CoA oxidation gives 90 ATP (1); activation costs 2, so net **120 ATP equivalents** (1). Check: $7(18) - 6 = 120$.

15. $v = 90(6)/(3+6) = 60$ µmol/min (1). $k_{cat} = 90/0.15 = 600\ \text{min}^{-1} = 10\ \text{s}^{-1}$ (2). $K_m = 0.003$ M, so $k_{cat}/K_m = 10/0.003 = 3.33 \times 10^3\ \text{M}^{-1}\text{s}^{-1}$ (1). $K_m = (k_{-1} + k_{cat})/k_1$ includes catalysis; it need not equal $K_d = k_{-1}/k_1$ (1).

16. Consistent with competitive inhibition in the simple model (1). α = 12/4 = 3, so $K_i = 6/(3-1) = 3$ mM (2). $v = 80(12)/(12+12) = 40$ µM/min (1).

17. Insulin RTK → IRS/PI3K/Akt promotes muscle GLUT4 translocation (2). Liver commonly uses GLUT2 and insulin instead regulates multiple hepatic metabolic processes; insulin-induced hepatic GLUT4 insertion is not the standard mechanism (1). Skeletal muscle lacks the principal direct glucagon responsiveness of liver; epinephrine, Ca^{2+}, and energy signals control its local glycogen mobilization (1).

18. Slow GTP hydrolysis prolongs active Gαs and adenylyl-cyclase stimulation, so ligand removal may not promptly end cAMP production (2). Phosphodiesterase-mediated cAMP degradation and phosphatases can reduce downstream signaling; receptor desensitization limits new receptor-driven input but may not switch off an already active mutant Gα (1). GEF activity exchanges GDP for GTP; GTPase activity hydrolyzes bound GTP to GDP (1).

19. GPCRs promote heterotrimeric G-protein nucleotide exchange; insulin receptor uses intrinsic tyrosine-kinase signaling in its preassembled receptor complex (2). PI3K generates PIP_3; PLC cleaves PIP_2 into IP_3 and DAG (1). IP_3 mobilizes Ca^{2+} from ER; DAG stays membrane-associated (1).

20. Evidence supports a short-term biomarker/pathway response, not demonstrated longer human lifespan or proof that AMPK mediates the benefit (2). An appropriate model-system AMPK loss-of-function plus rescue experiment can test necessity/causal involvement (1). Include suitable diet/intake controls and nutritional adequacy; for a drug study include vehicle and consider intake matching to separate direct effects from appetite changes (1). Confounding and healthspan versus lifespan remain distinct issues.

21. Fatty-acid oxidation supplies liver ATP/NADH and acetyl-CoA (1). ATP supports gluconeogenesis and acetyl-CoA activates pyruvate carboxylase (1). OAA availability/diversion and abundant acetyl-CoA favor ketogenesis under the relevant fasting conditions (1). Even-chain fatty acids generally supply energy, not net glucose carbon; glycerol from triacylglycerol and odd-chain propionyl carbon are distinct exceptions/routes (1).

22. Transamination transfers an amino group without directly releasing free ammonia (1). Oxidative deamination can release ammonium from glutamate (1). Glutamine and alanine carry nitrogen (1); liver synthesizes urea for renal excretion (1).

23. A high-dose hook/prozone-type effect is plausible in an appropriate sandwich assay (1); serial dilution and controls can test whether the signal returns to the validated range (1). Other matrix/interference explanations also require consideration (1). Calibration/range validation is necessary because signal is not automatically linear or uniquely proportional to concentration over every possible value (1).

24. F2,6BP falls (1); PFK-1 activation falls and FBPase-1 inhibition is relieved (1); this favors less glycolysis and more gluconeogenesis in the stated liver setting (1). The bifunctional enzyme has tissue-specific isoforms/regulation, so the exact hormonal/phosphorylation logic is not universal (1).

Interpreting your practice result

Use errors by mechanism rather than total score alone. If a calculation is wrong, classify the cause as setup, equation, units, stoichiometry, or arithmetic. If a mechanism answer is weak, rewrite it as a causal chain and retest it the next day. A strong mark on this original practice set does not establish the difficulty or coverage of the actual examination.

21. Revision schedule and final readiness check

A 14-session plan

This is a flexible sequence, not a calendar tied to an assumed exam date. For each session use roughly half the time for explanation/problem solving, one quarter for drawing from memory, and one quarter for cumulative recall. Adjust duration to your available time.

Session	Main work	Closed-note output
1	Sections 1–2: buffers, charge, pI	Two buffer problems; one charge/pI problem
2	Sections 3–4: folding, Hb/Mb	Structural-level sketches; binding-curve explanation
3	Sections 5–6: antibodies, sugars	Label IgG; draw glucose anomers and three disaccharide linkages
4	Sections 7–8: membranes, transport	Transport comparison; signed electrochemical calculation
5	Section 13: kinetics	Estimate K_m/V_{max}; calculate turnover; classify inhibition
6	Section 9: glycolysis and fermentation	Ten glycolytic steps, controls, and net ledger
7	Section 9: PDH and TCA	Cycle intermediates, carbon counts, cofactors, products
8	Section 10: gluconeogenesis	Four bypass enzymes; energy cost; Cori-cycle reasoning
9	Section 11: oxidative phosphorylation	Electron/proton map and inhibitor predictions
10	Section 12: fatty acid oxidation	Shuttle/cycle; palmitate and C18 yield calculations
11	Sections 14, 16–17: hormones and receptors	Gs/Gi/Gq; insulin/Akt; fed/fasted tissue table
12	Sections 15 and 18: aging and integration	Evidence-based CR paragraph; integrated fasting explanation
13	Section 19 practice paper	Complete under the suggested conditions, then mark
14	Targeted repair and cumulative recall	Redo missed questions without answers; check formula-sheet use

If time is short, combine adjacent content sessions but preserve a timed practice attempt and a separate error-correction session. Do not omit an official topic merely because it is brief in the supplied focused notes.

Readiness checklist

Without notes, can you:

- Calculate a buffer before/after strong-acid addition and explain capacity?
- Estimate peptide charge and identify the correct pI-bracketing titrations?
- Draw/describe protein structural levels and predict a mutation’s effect?
- Explain Hb cooperativity, the Bohr effect, and affinity versus capacity?
- Label Fab/Fc and distinguish antigen/epitope, affinity/avidity?
- Draw glucose anomers and identify reducing versus nonreducing linkages?
- Predict fluidity changes and distinguish passive, primary, and secondary transport?
- Calculate a signed electrochemical free energy with units?
- Reconstruct glycolysis, PDH, and TCA carbon/electron/ATP accounting?
- Name gluconeogenic bypasses and explain energy cost and carbon restrictions?
- Trace electron entry, proton pumping, and ATP coupling, including ETF versus complex II?
- Predict inhibitor/uncoupler responses and calculate fatty-acid yields?
- Solve Michaelis–Menten/inhibition problems without equating K_m automatically with K_d?
- Explain fed/fasted responses by tissue and distinguish Gs, Gi, Gq, RTK, and Ras?
- Discuss CR mechanisms while distinguishing model evidence, biomarkers, healthspan, and lifespan?

Using the companion formulas-only sheet

The separate PDF is **two A4 pages intended for duplex printing onto one sheet**. It contains mathematical relations without pathway diagrams, narrative definitions, topic headings, worked examples, or a reaction-enzyme list. This conservative layout follows the supplied outline's formulas-only requirement; exam administrators determine admissibility.

Learn the symbols and assumptions here before relying on that sheet. Concentrations in ratios must refer to the appropriate free species or model quantities. Use kelvin for T; $R = 8.314$ J mol^{-1} K^{-1} or 0.008314 kJ mol^{-1} K^{-1}, not a mixture. $F = 96485$ C mol^{-1}. Base-10 logs appear in pH equations; natural logs appear in free-energy equations. Define transport direction before using electrical potential differences. ATP yield formulas assume the stated P/O convention and substrate class.

For the β-oxidation shortcut, n is the even number of carbons in a free saturated fatty acid. For the two glucose-yield expressions on the sheet, 8 matrix NADH, 2 cytosolic NADH equivalents, 2 FAD-linked pairs, and 4 direct ATP equivalents are being counted; the 30/32 difference represents the shuttle convention. These are bookkeeping models, not exact measured ATP yields for every cell.

Source register

[O] Exam scope and rules. *05_Biochemistry-BMEB_2026.docx*, supplied by the user. Identifies BMED 795, Biochemistry, Biomedical Engineering and Biotechnology; lists 17 topics, their primary-textbook chapter assignments, and permitted exam materials. No exam date, marks distribution, official question format, or past questions were supplied.

[F] Supplied revision material. *Biochemistry_WQE_Focused_Study_Guide.pdf*, 58 pages, supplied by the user. Page references throughout this guide refer to that document. Foundations pp. 3–9; amino acids/proteins pp. 10–17; enzymes/energy pp. 18–26; carbohydrate chemistry pp. 27–34; lipids/membranes/transport pp. 35–41; laboratory concepts pp. 42–44; glycolysis/fermentation/glycogen pp. 45–48; PDH/TCA pp. 49–50; oxidative phosphorylation pp. 51–53; fatty acid oxidation pp. 54–55; nitrogen/integration pp. 56–58.

[T, not accessed] Uploaded primary textbook. *Principles of Biochemistry 5e By Robert Horton.pdf*. Transfer blocked by the 32 MiB per-file limit. The primary-textbook bibliographic identification in [O] is Moran, Horton, Scrimgeour, and Perry, *Principles of Biochemistry*, 5th ed., Pearson, 2011. The outline also lists Voet and Voet, *Biochemistry*, 4th ed., Wiley, 2010, as reference material; that book was not supplied or consulted. Chapter labels in this guide come from [O].

New material. Topic expansions, study priorities, original calculations, practice questions/answers, and the revision schedule were written for this guide using standard biochemistry knowledge. They are educational aids, not quotations, official exam predictions, or claims of line-by-line textbook verification.